Magic in particle systems

Márton Balázs

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Stochastics and Influences, Erdős Center
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Happy birthday Bálint!

The models

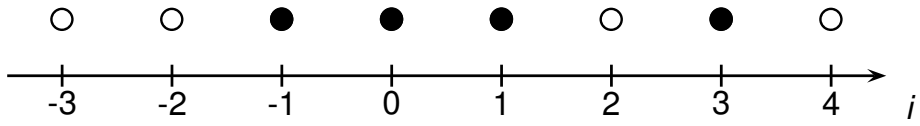
- Asymmetric simple exclusion process
- Zero range
- Bricklayers

Current

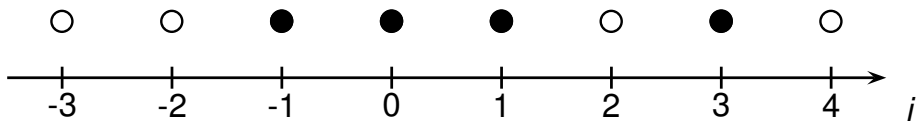
Martingales

EJS

Asymmetric simple exclusion



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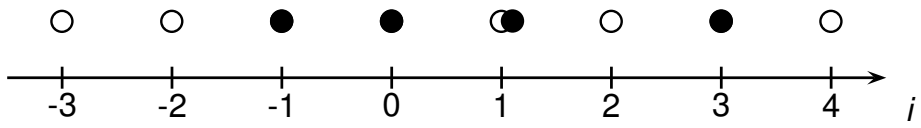


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to the right with rate p,
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Asymmetric simple exclusion

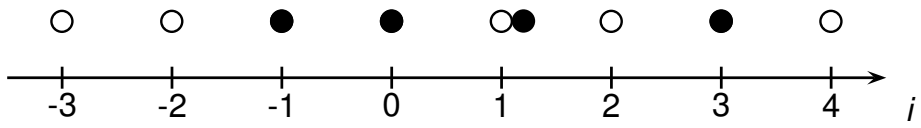


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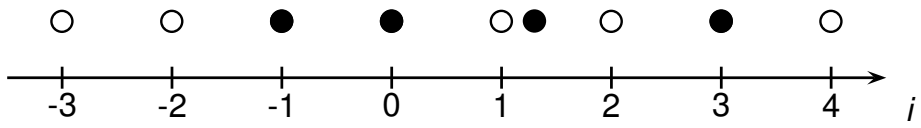


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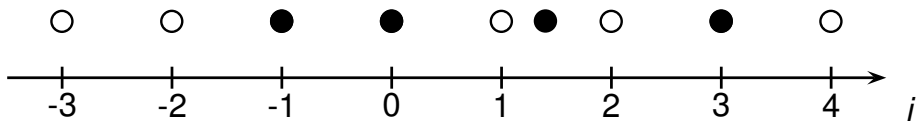


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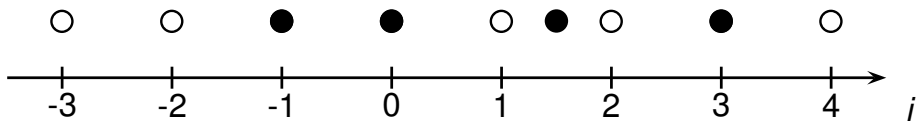


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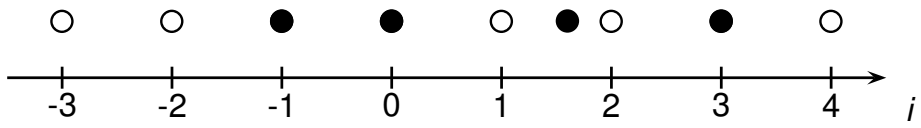


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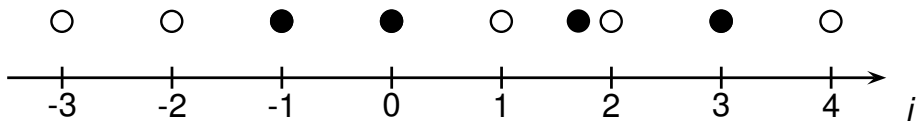


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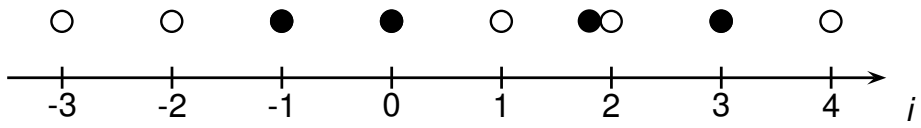


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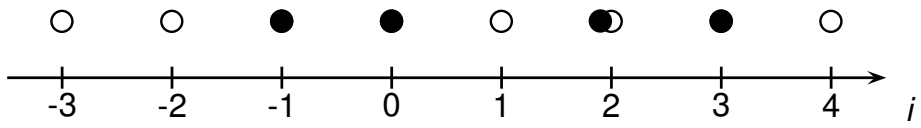


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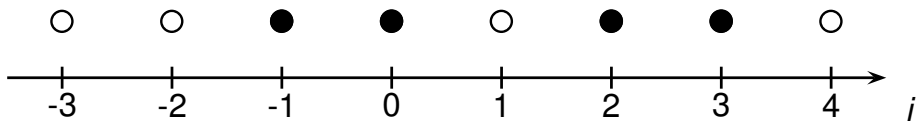


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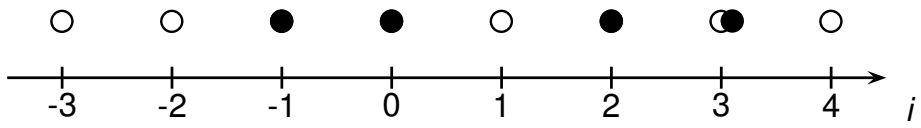


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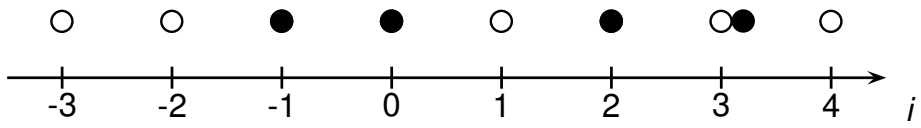


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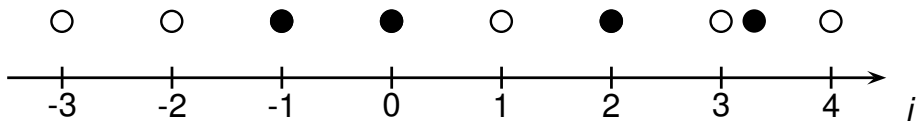


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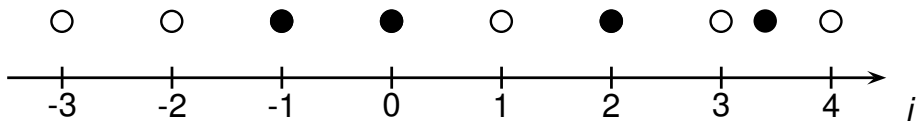


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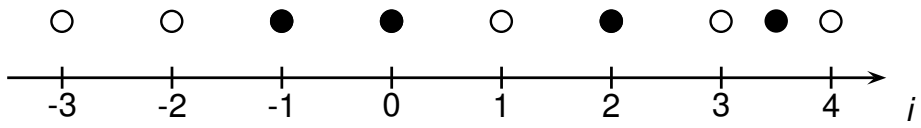


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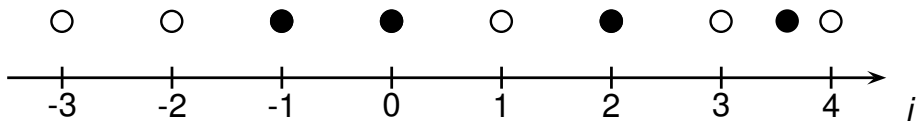


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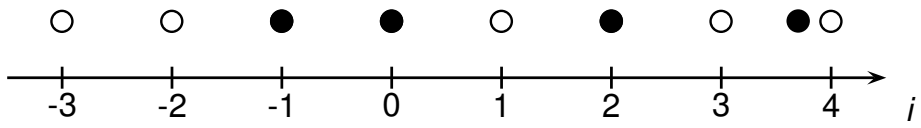


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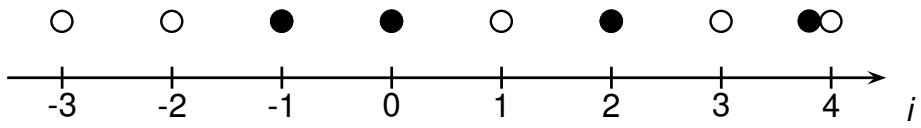


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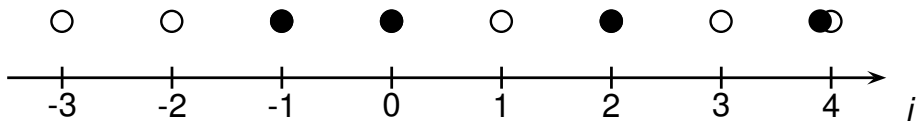


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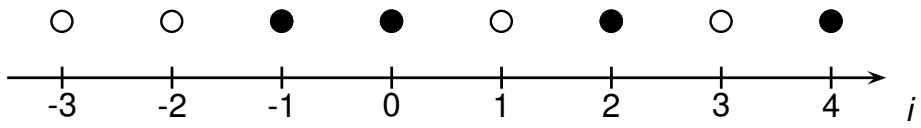


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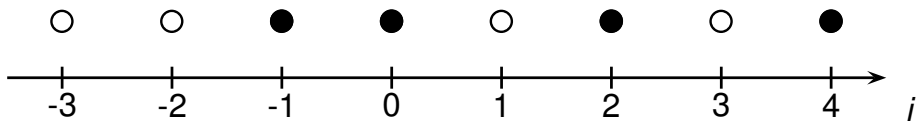


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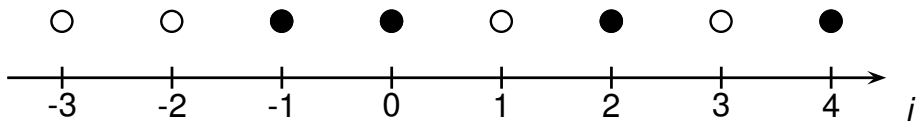


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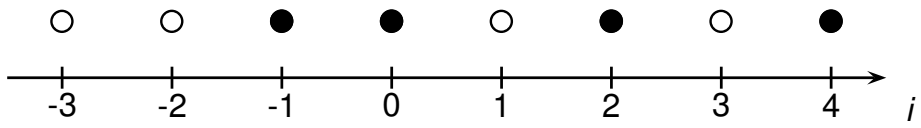


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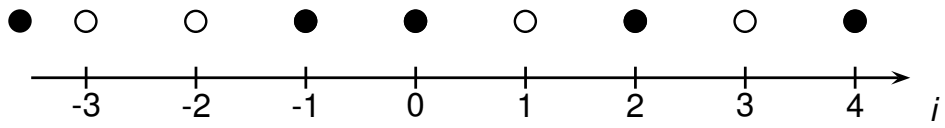


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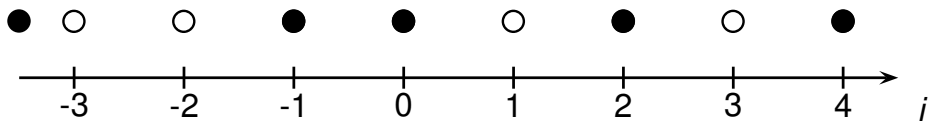


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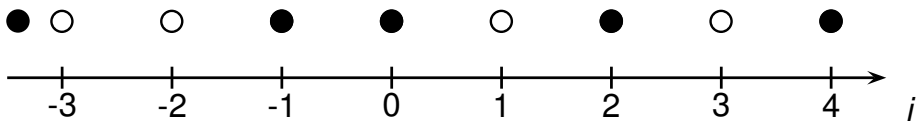


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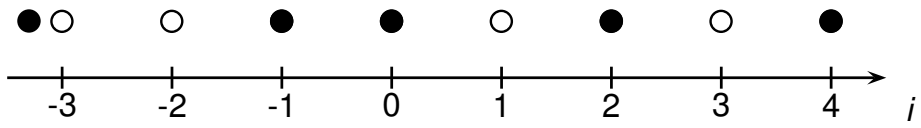


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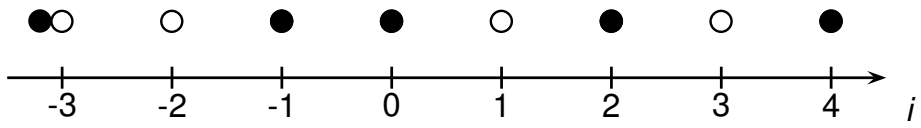


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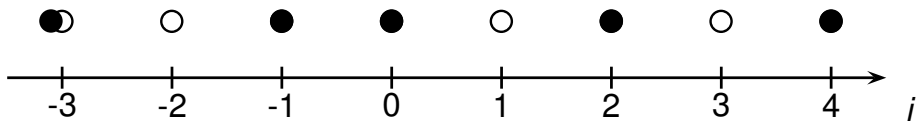


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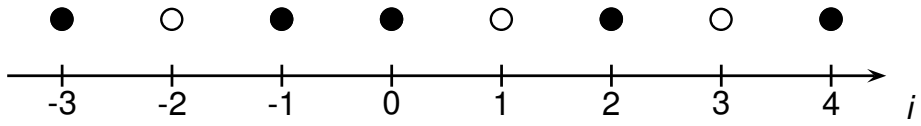


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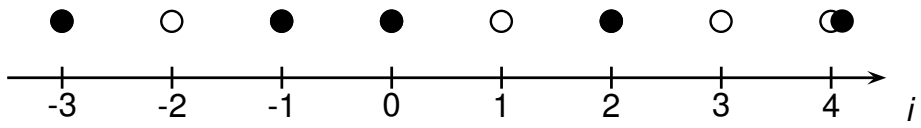


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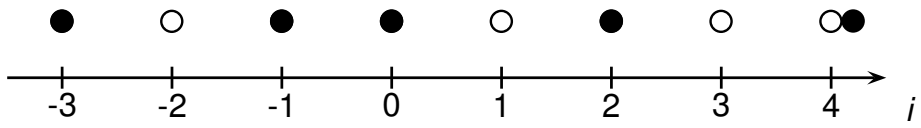


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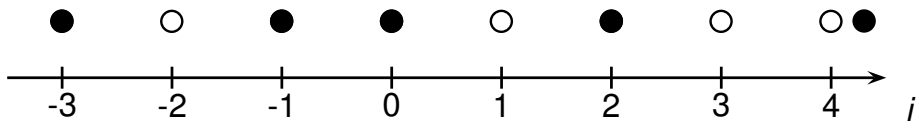


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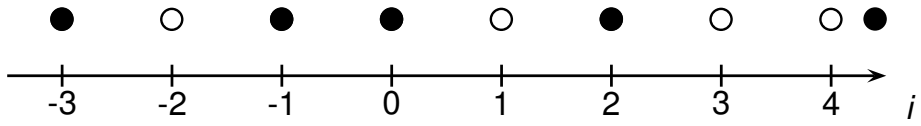


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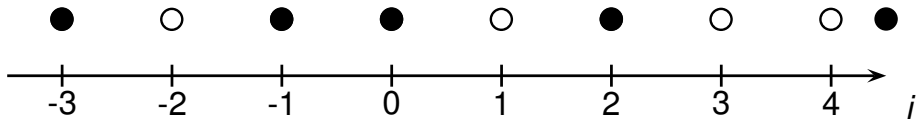


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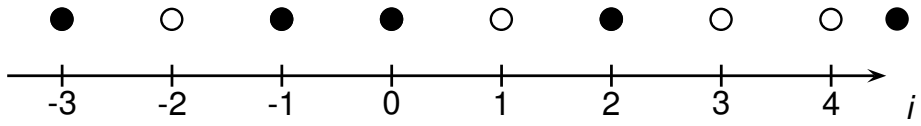


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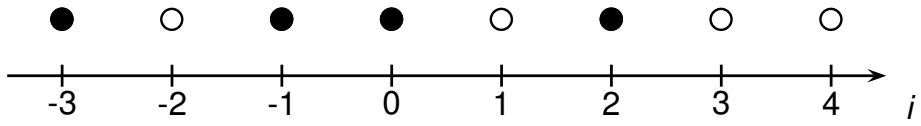


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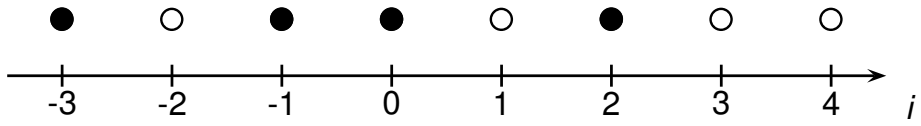


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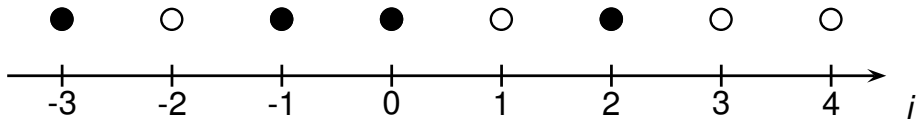


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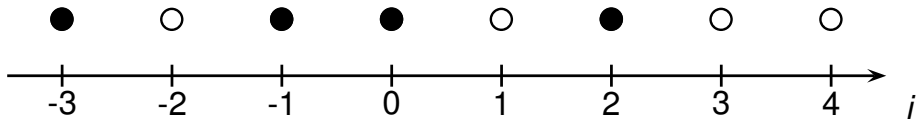


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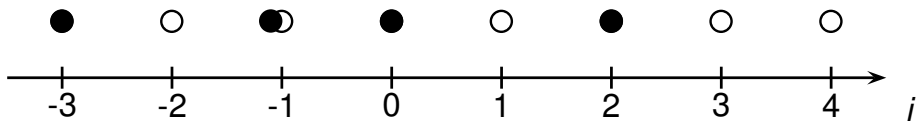


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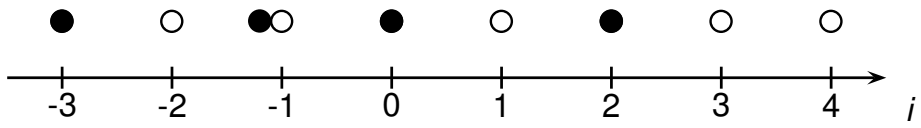


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to the left with rate $q = 1 - p < p$.

The jump is suppressed if the destination site is occupied by another particle.

Asymmetric simple exclusion

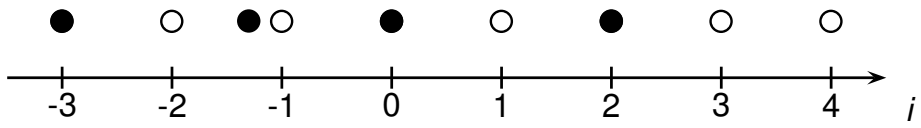


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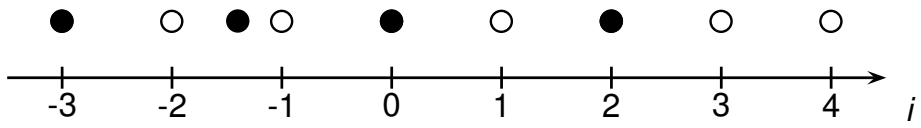


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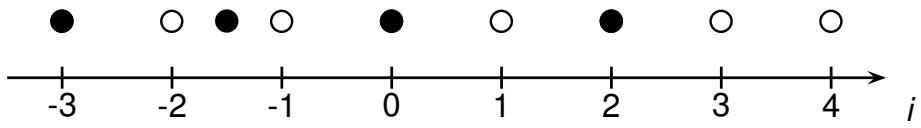


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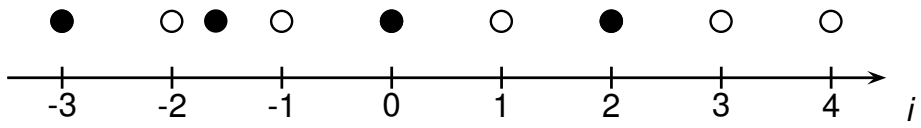


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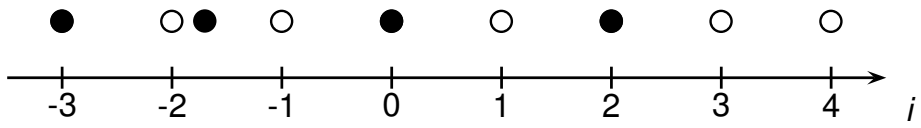


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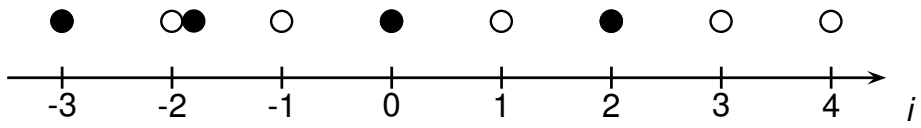


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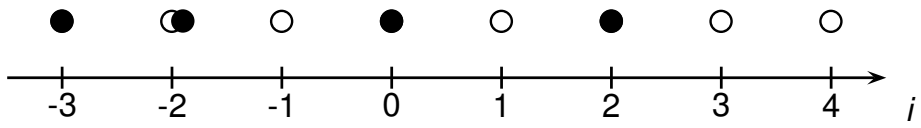


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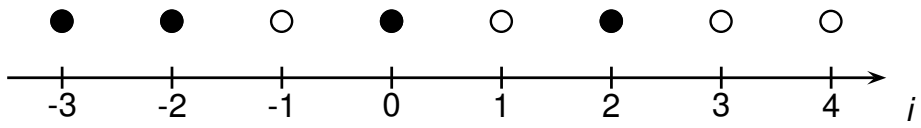


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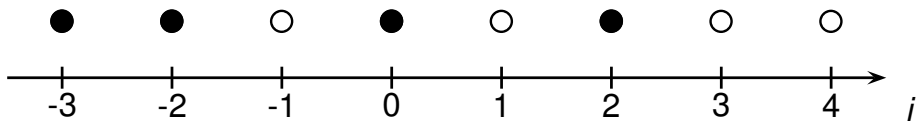


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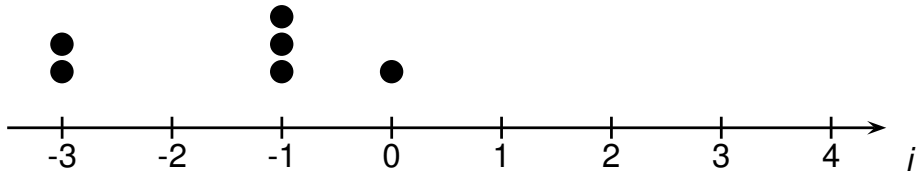
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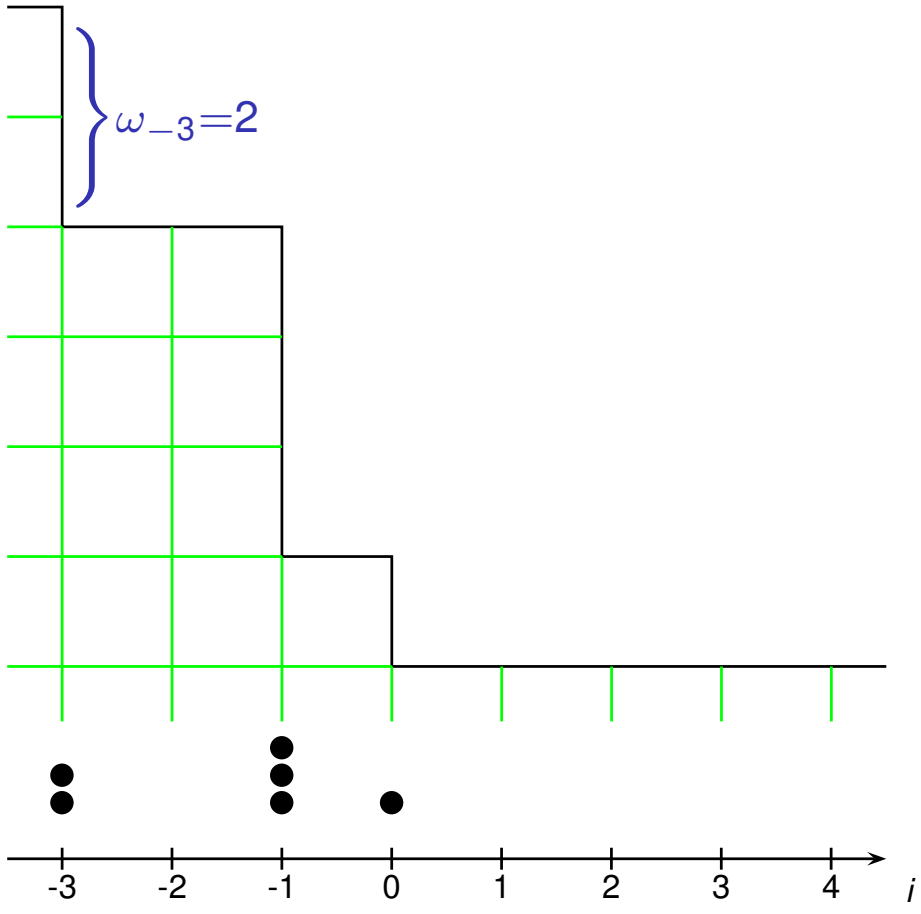
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I.i.d. Bernoulli(ϱ) distribution is time-stationary for any ($0 \leq \varrho \leq 1$).

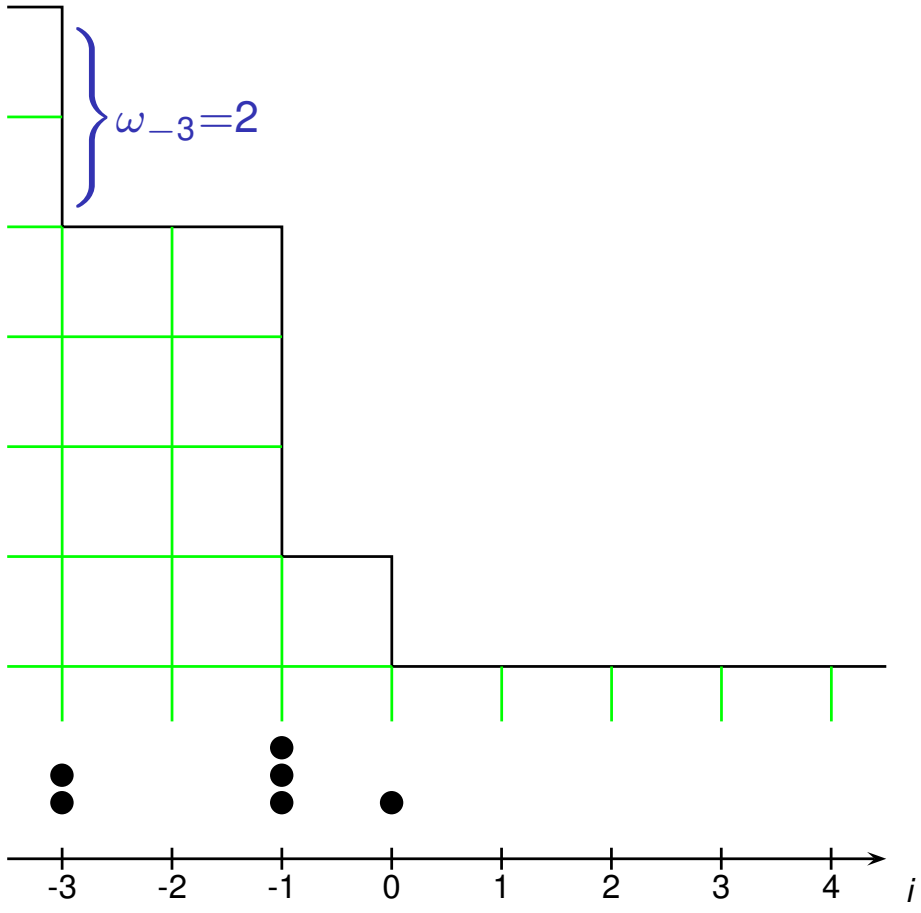
The asymmetric zero range process



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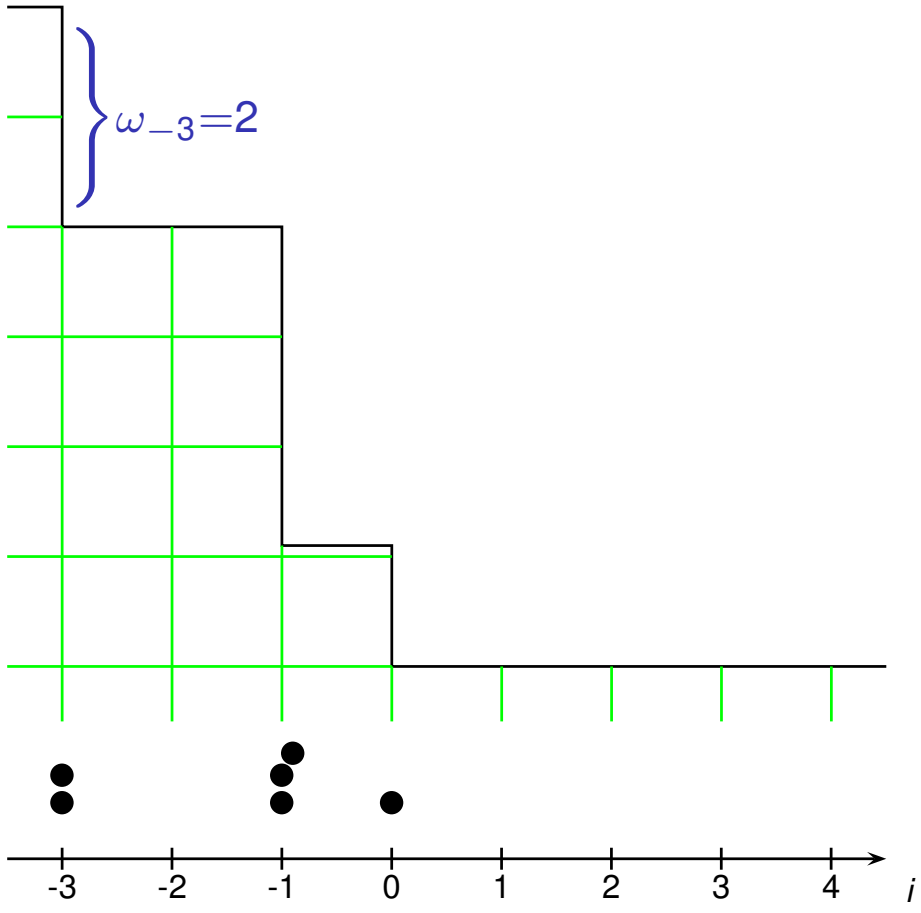


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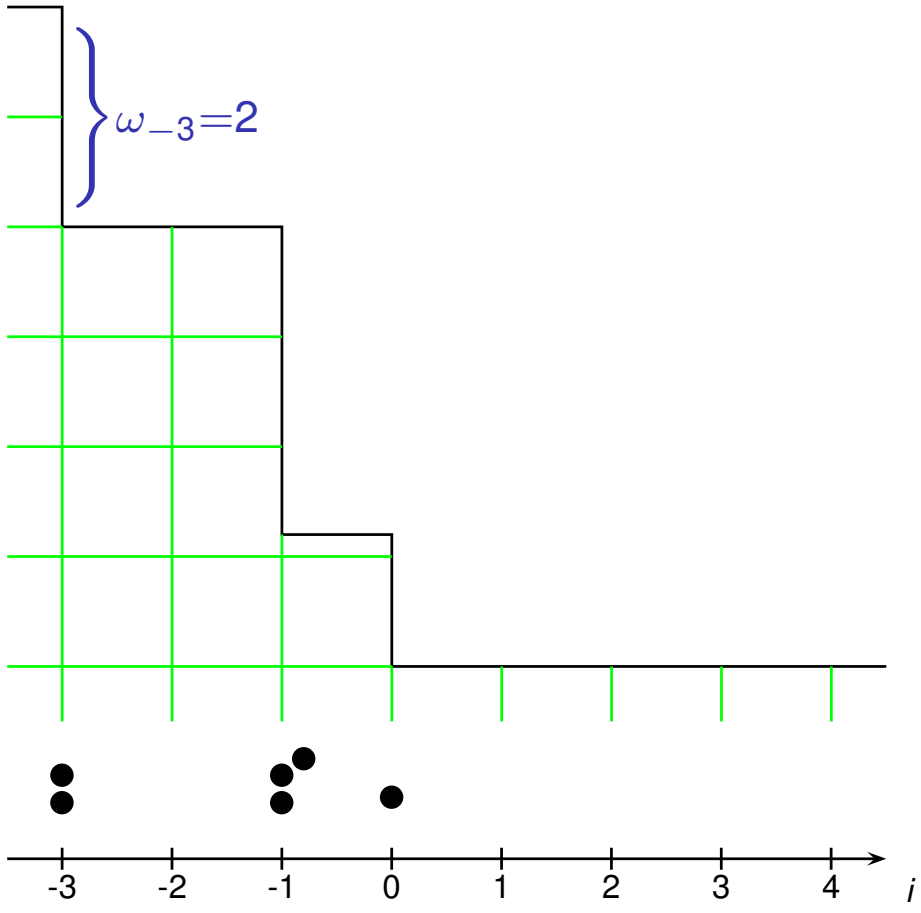
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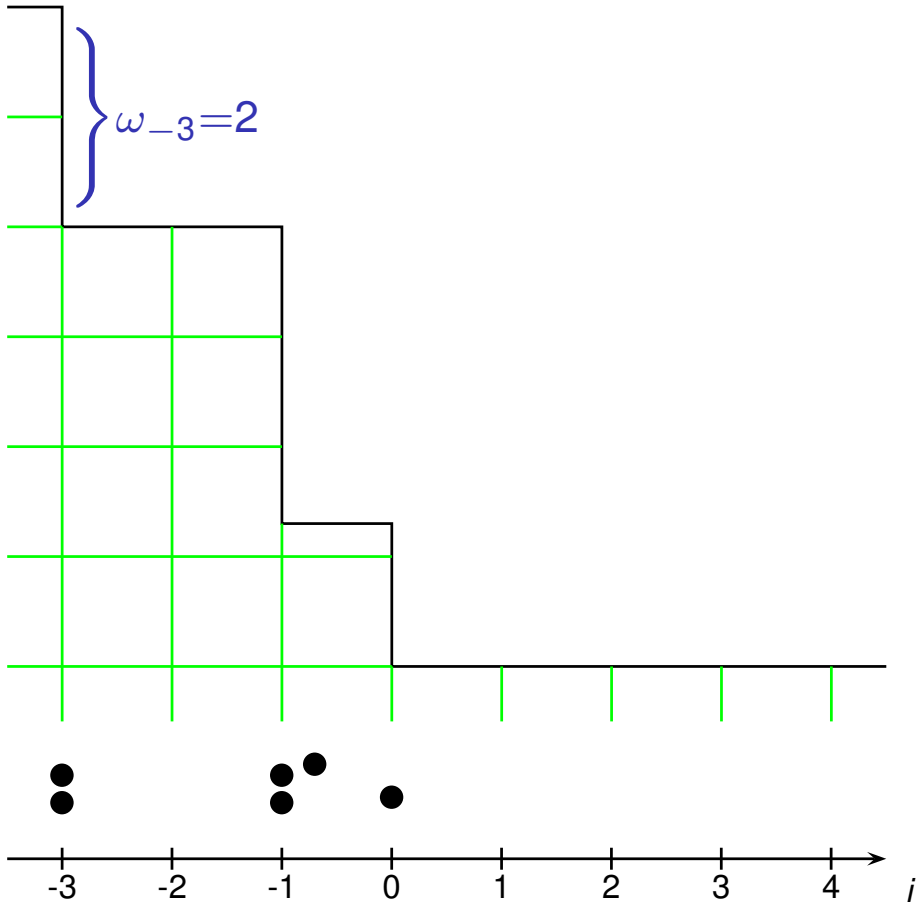
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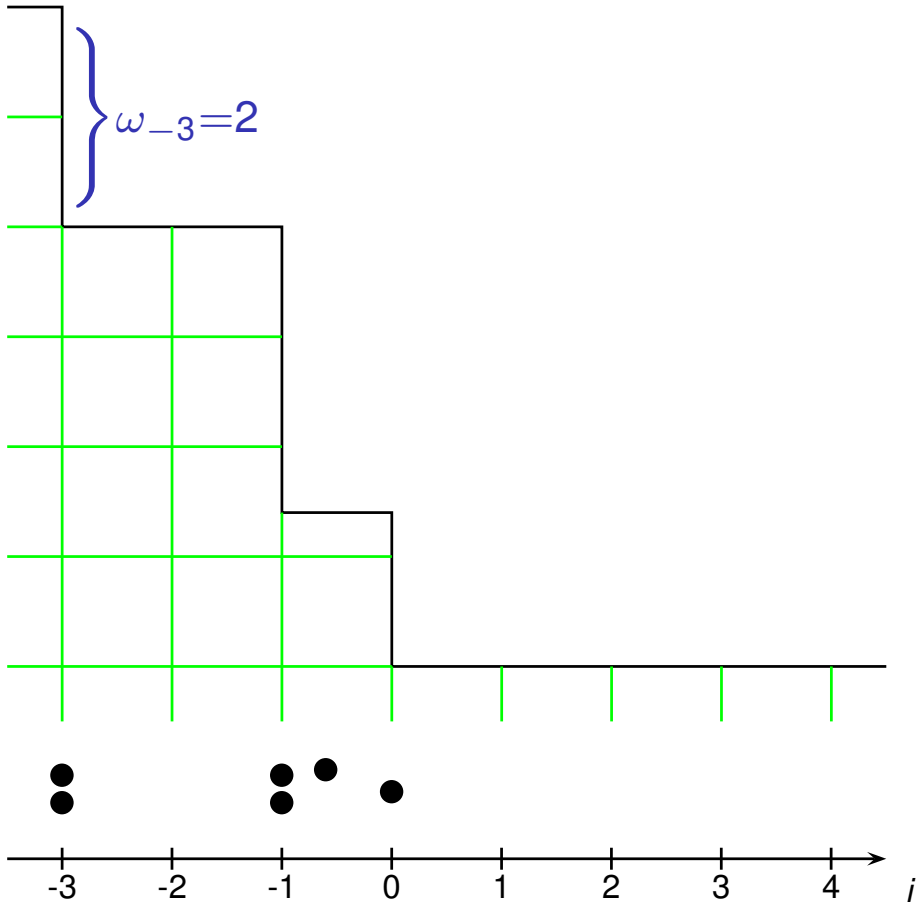
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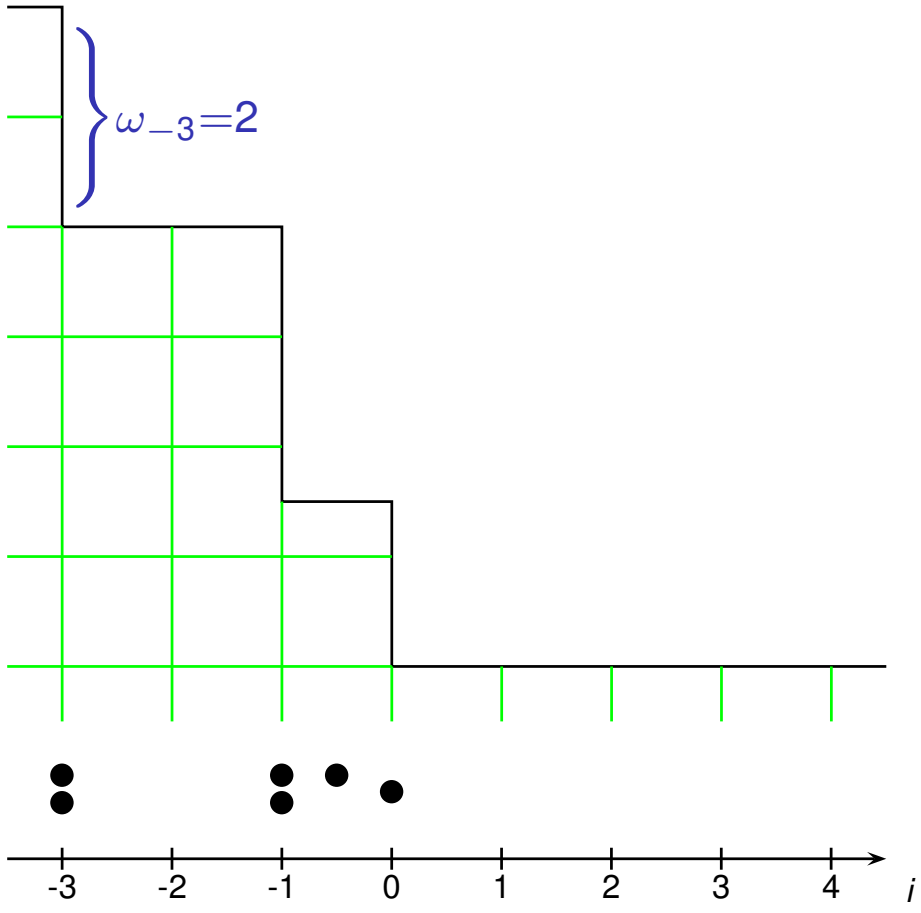
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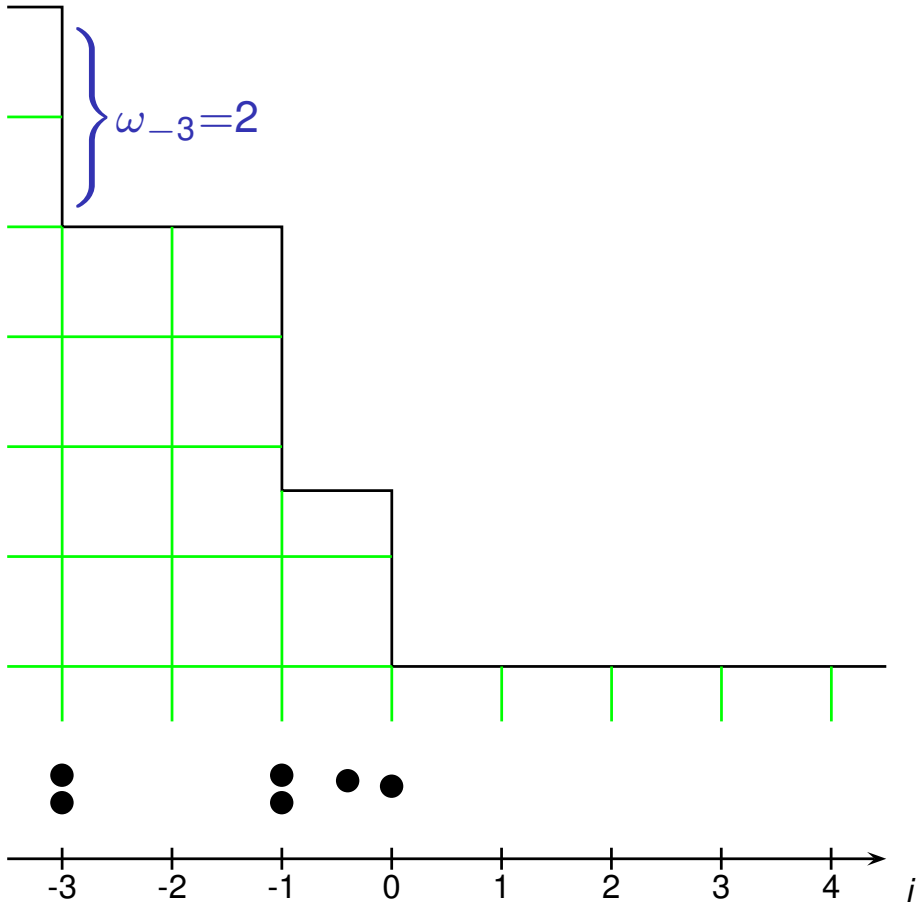
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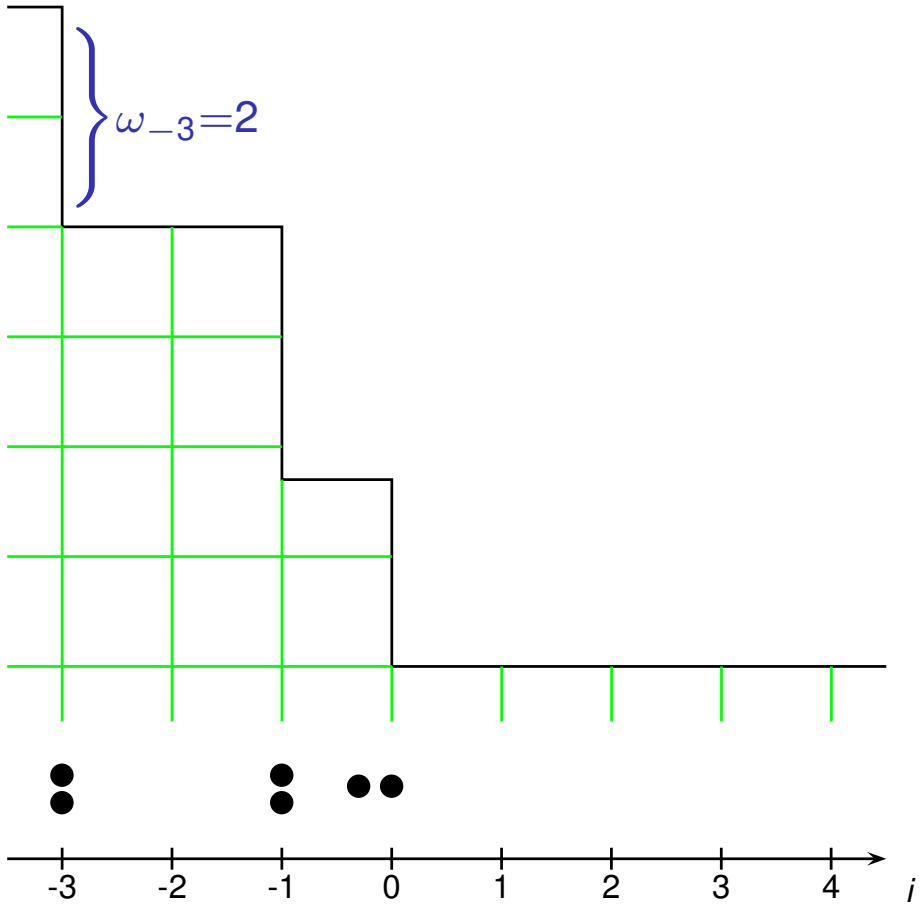
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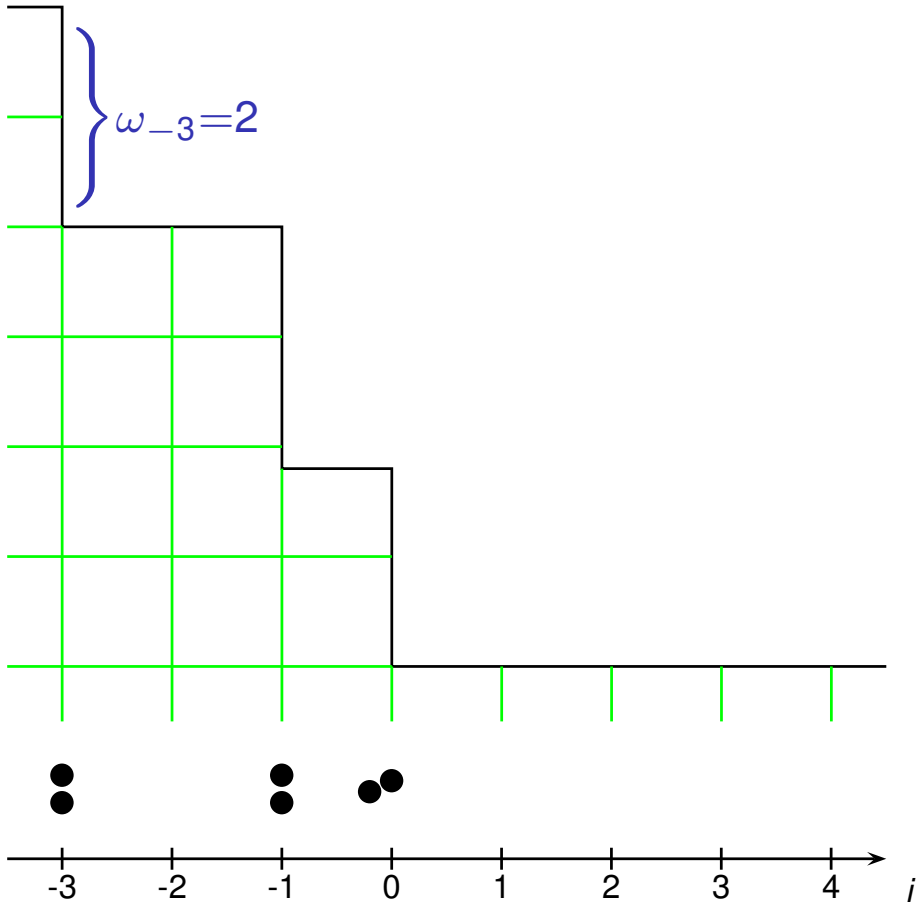
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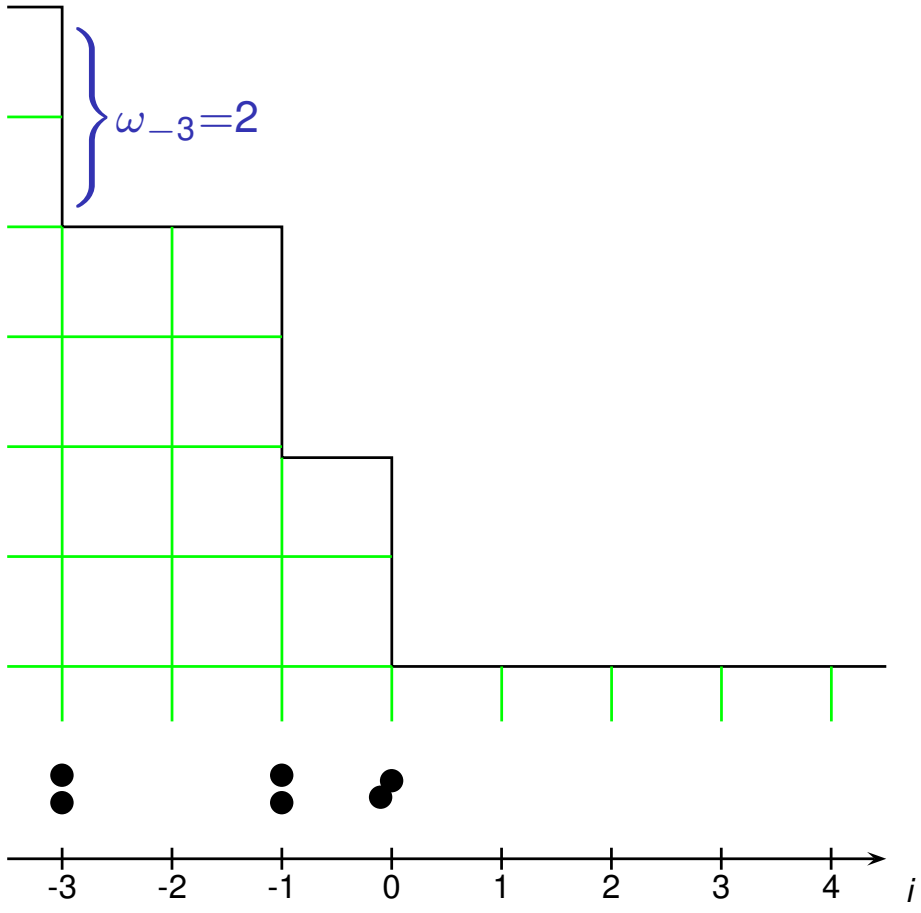
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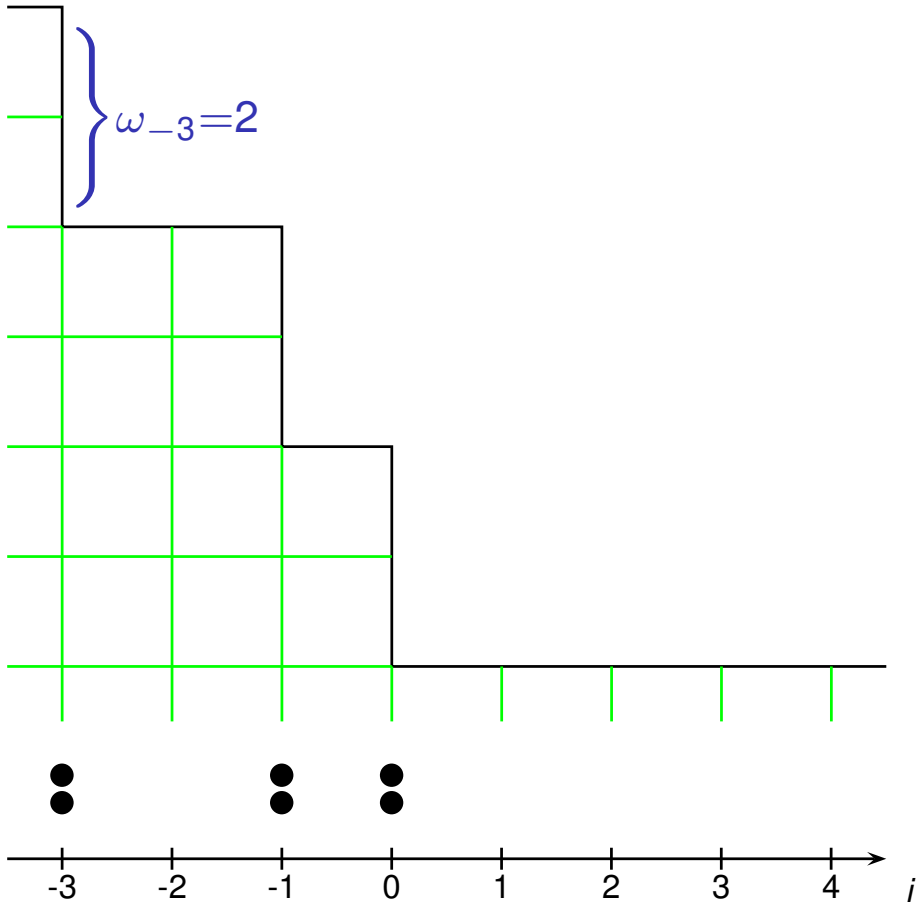


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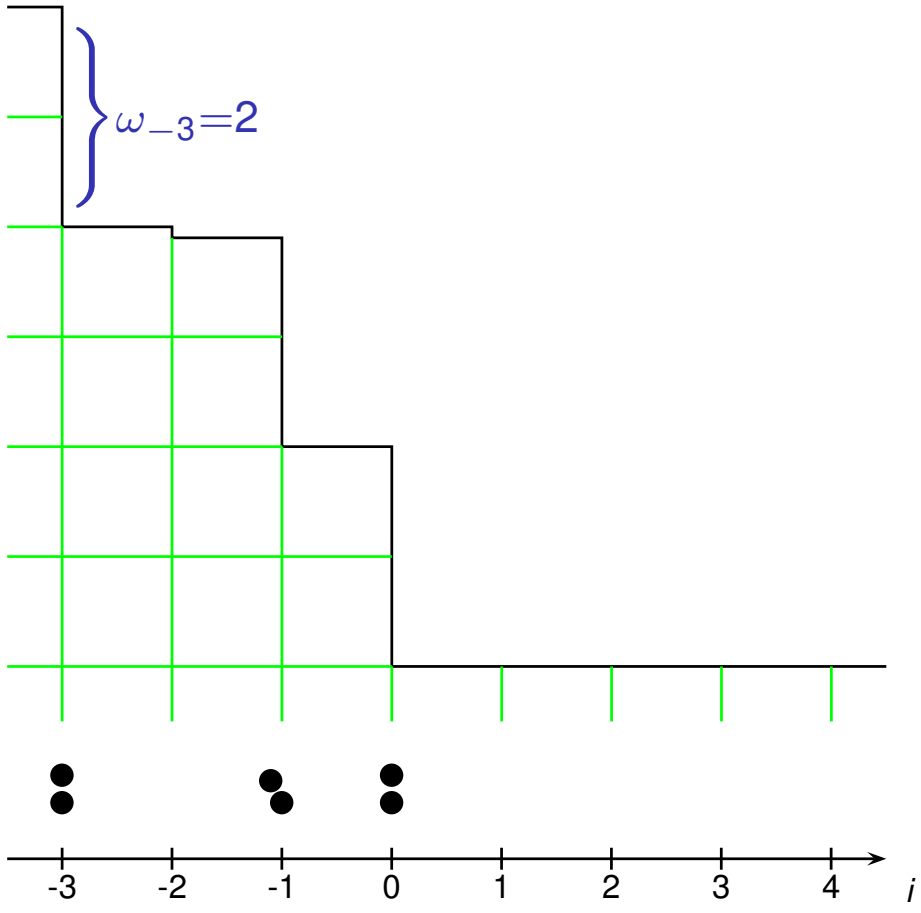
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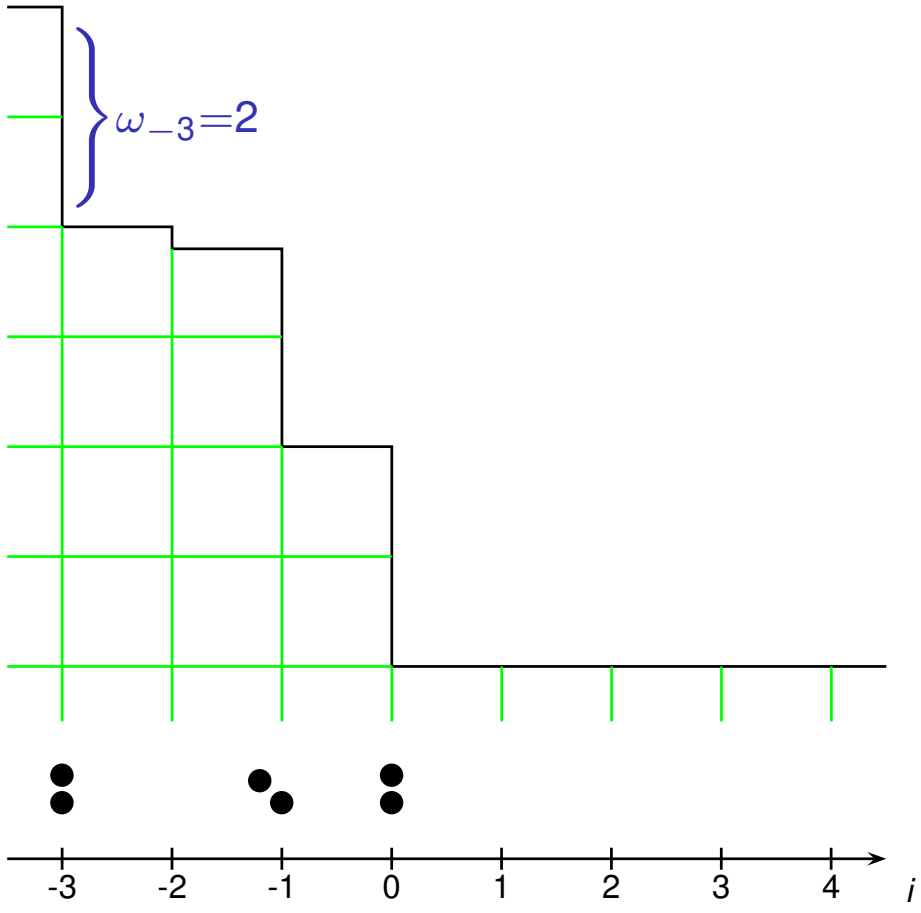
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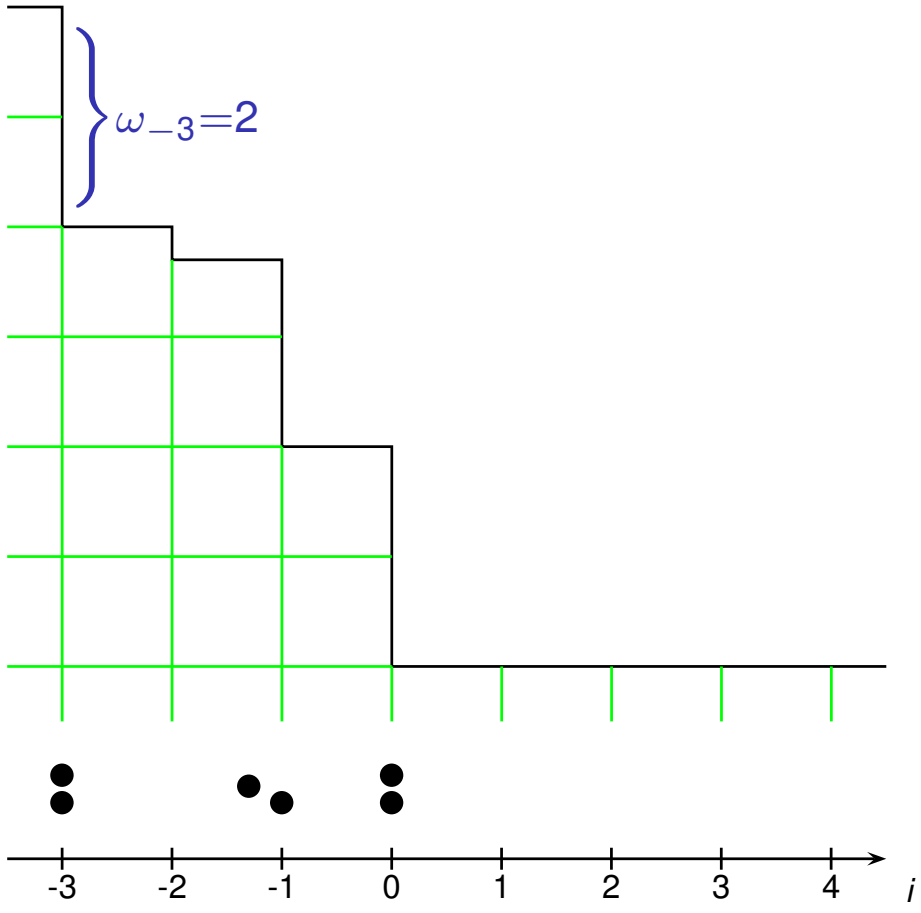
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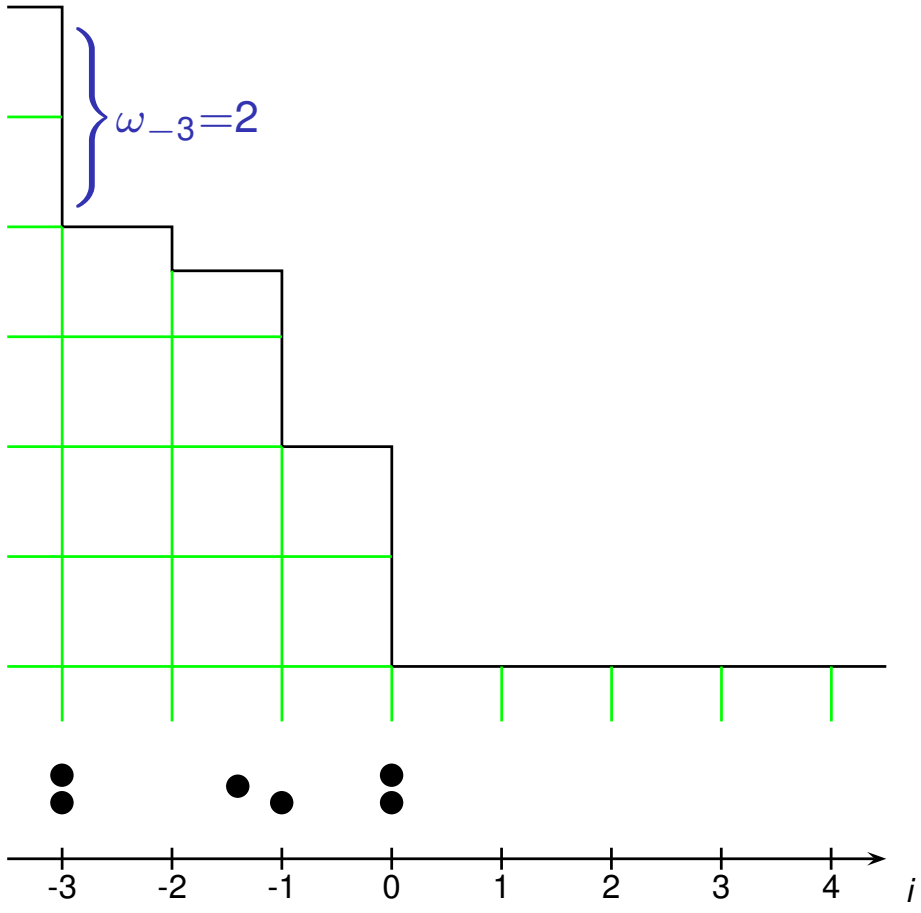
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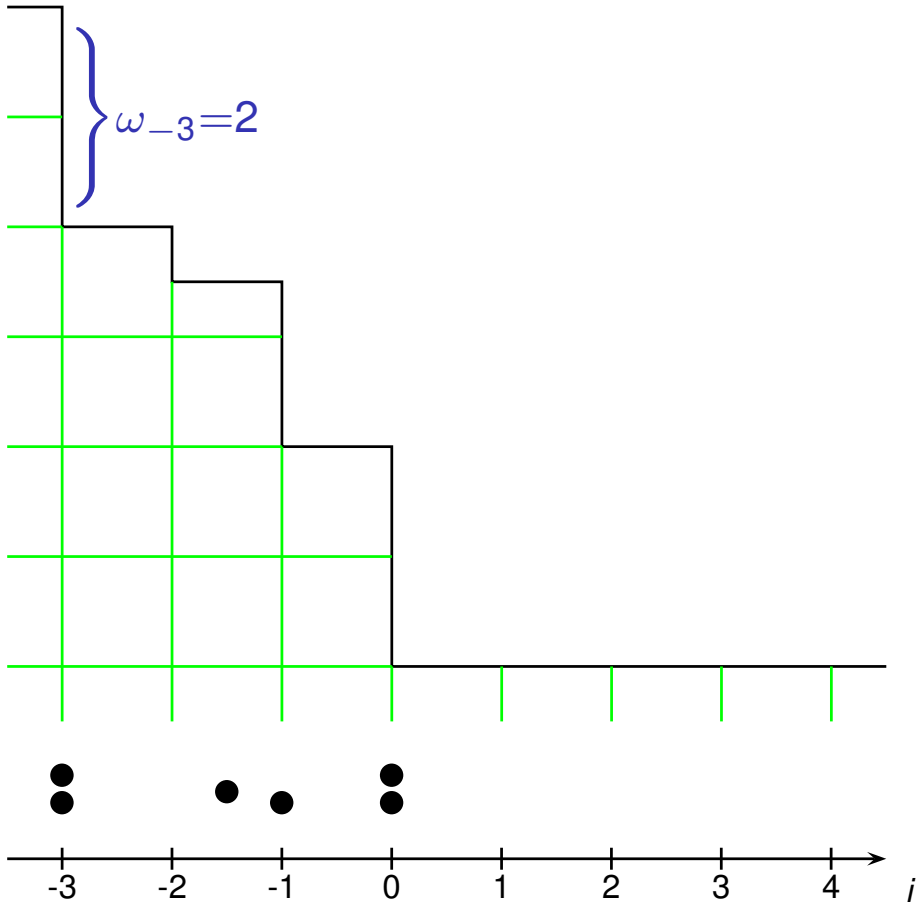
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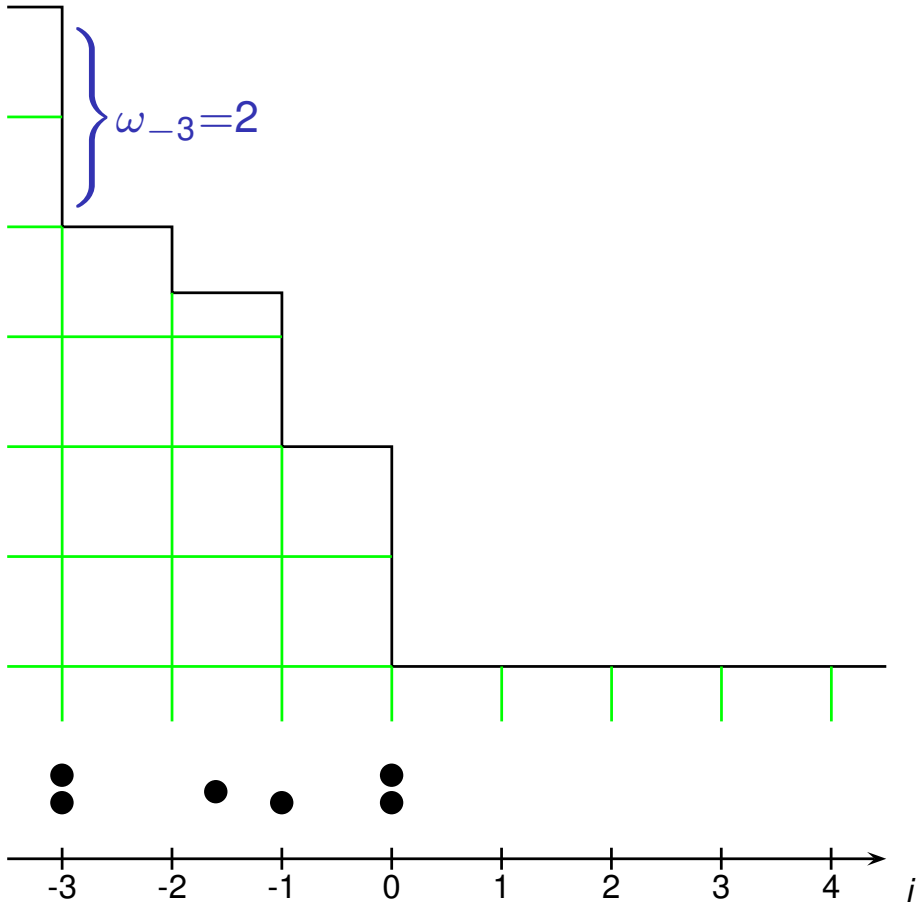
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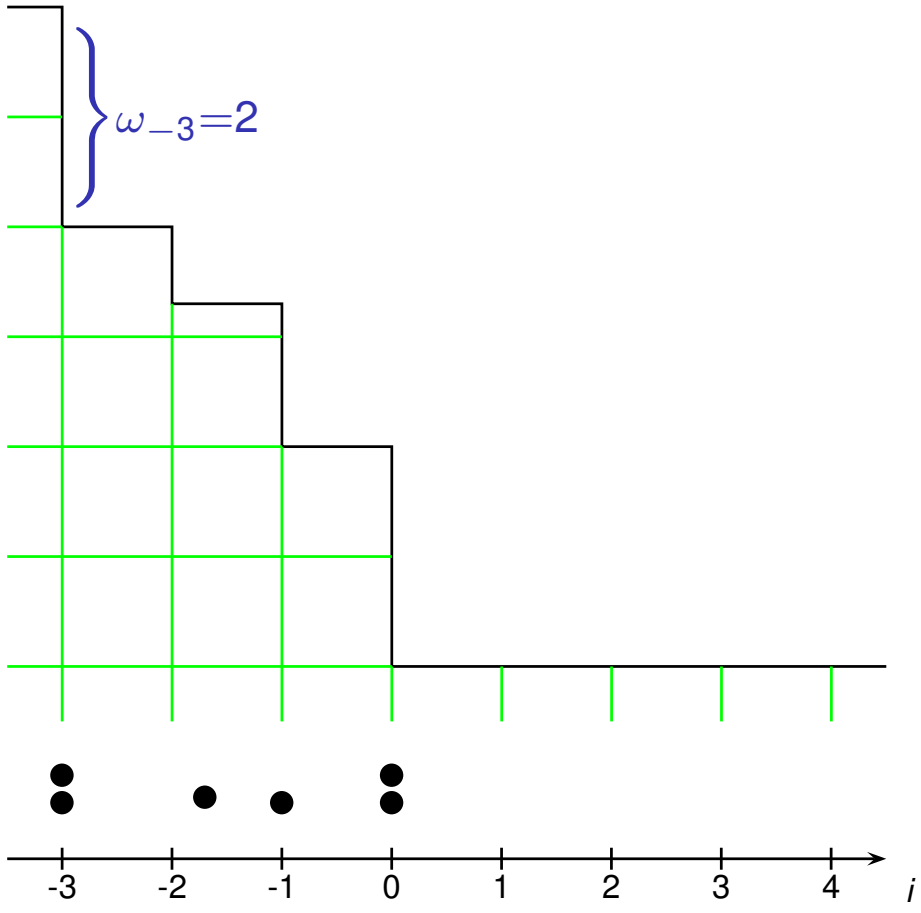
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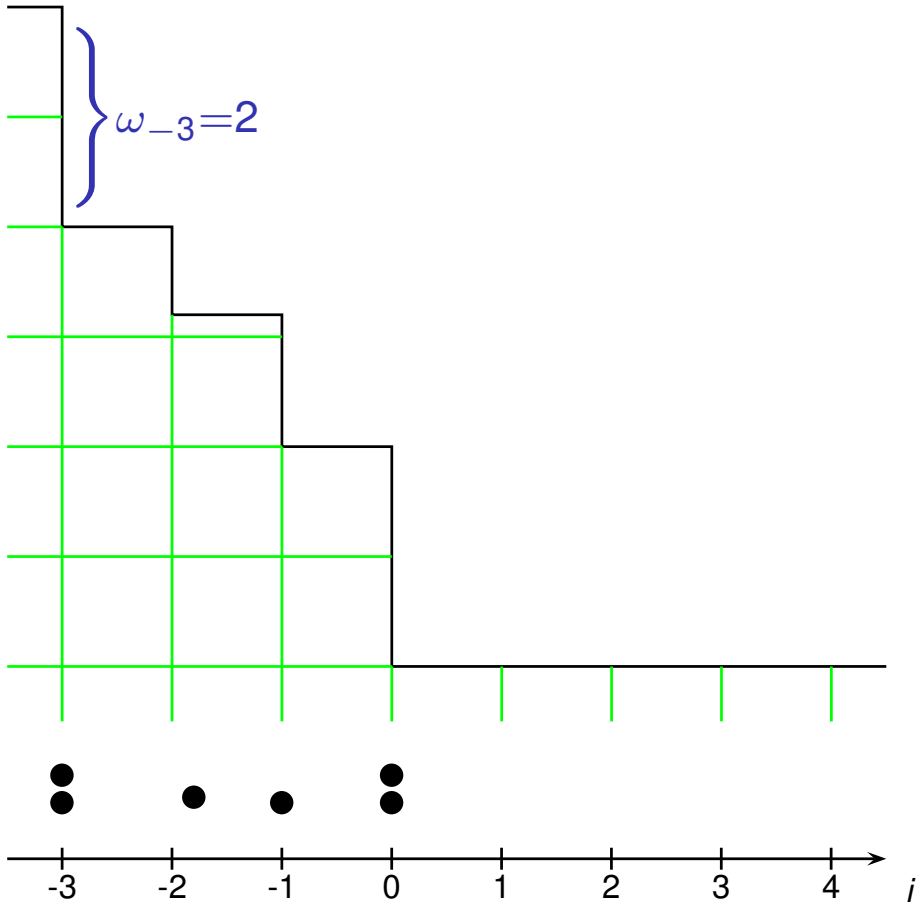
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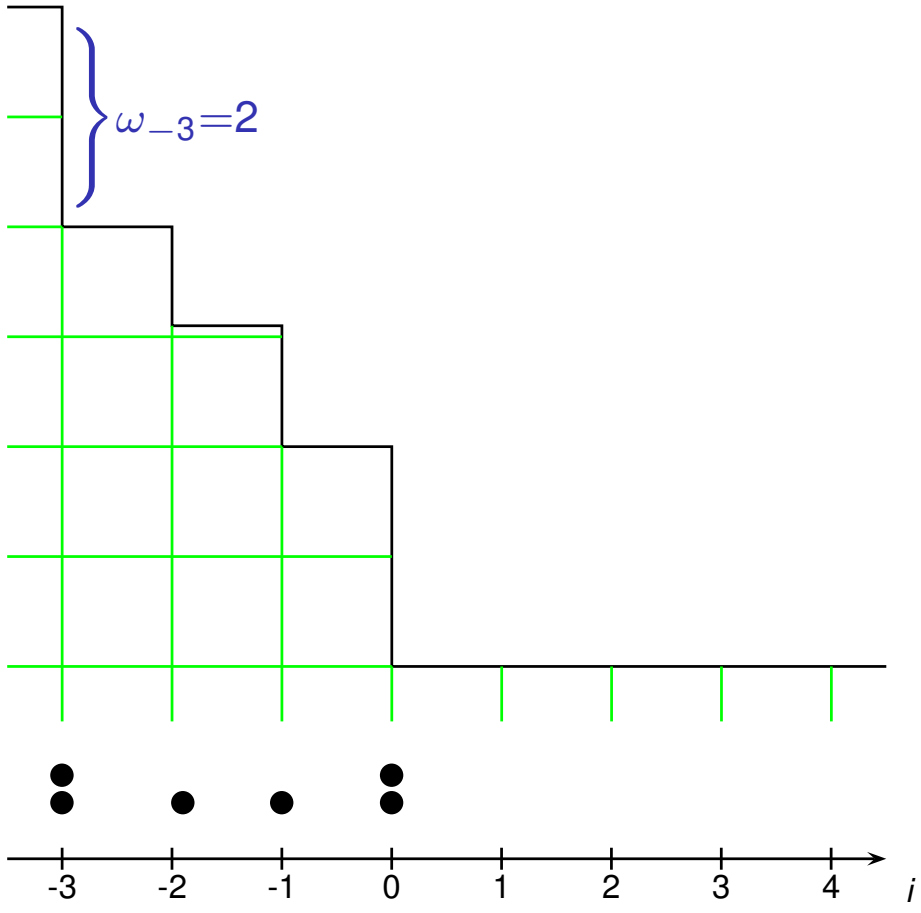
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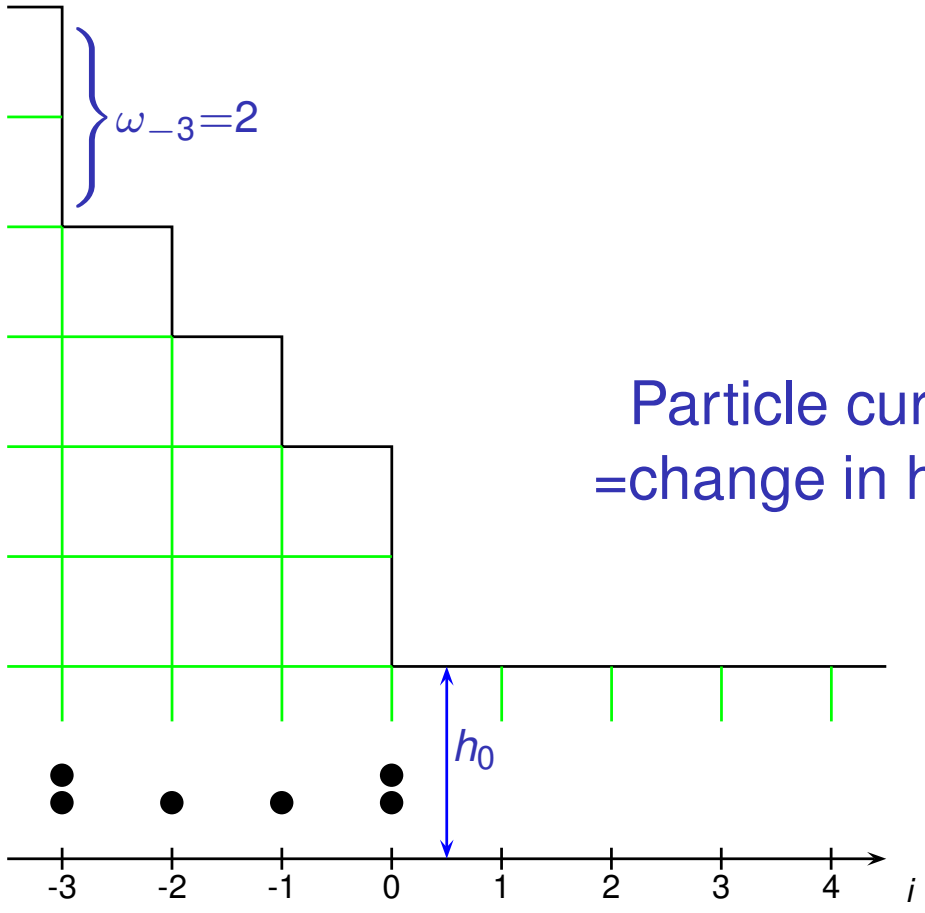
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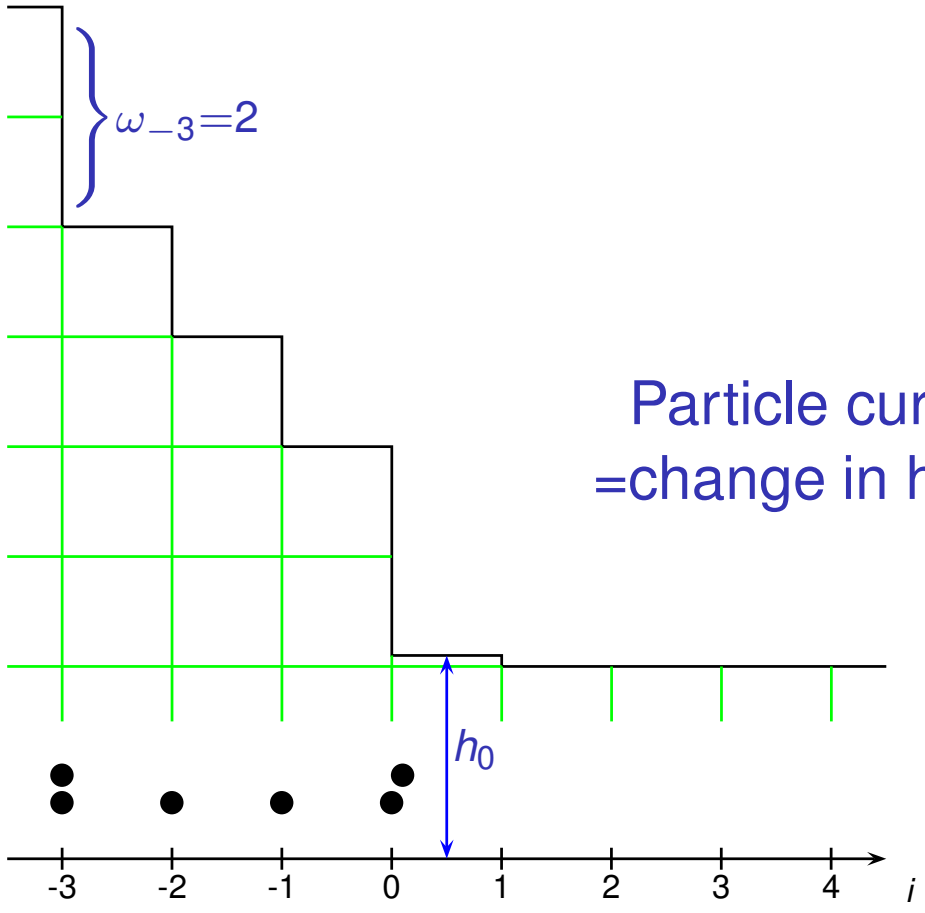
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Particle current
=change in height

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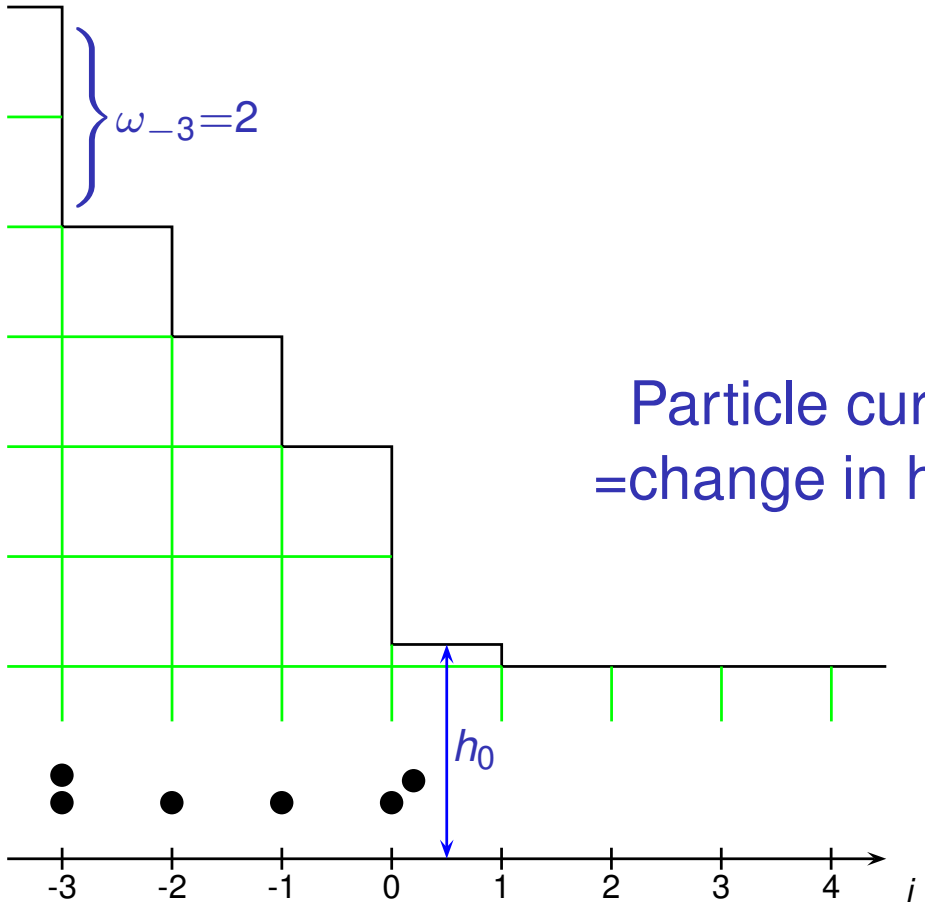
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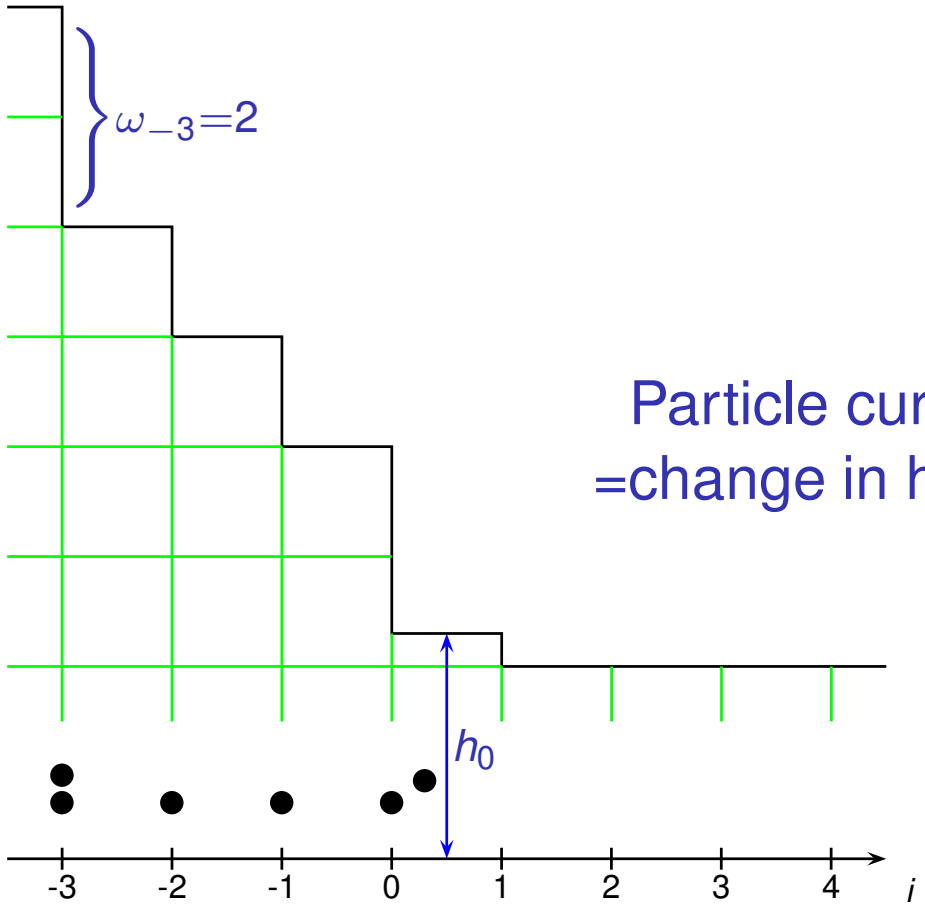
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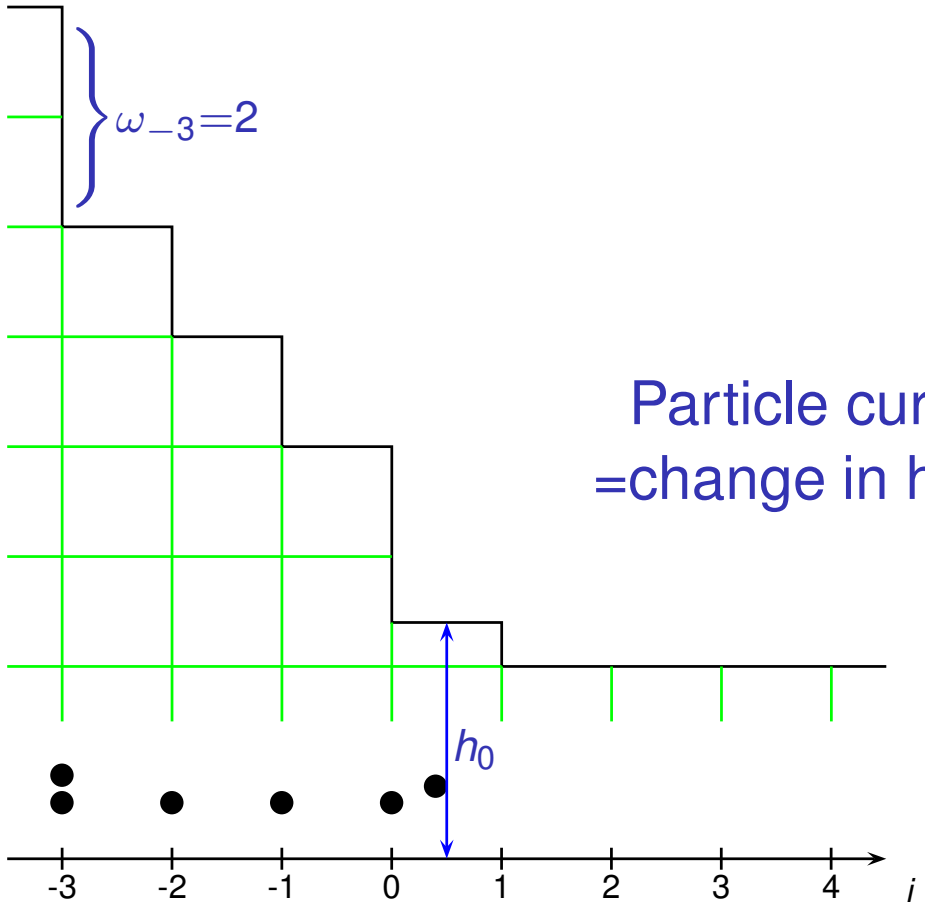
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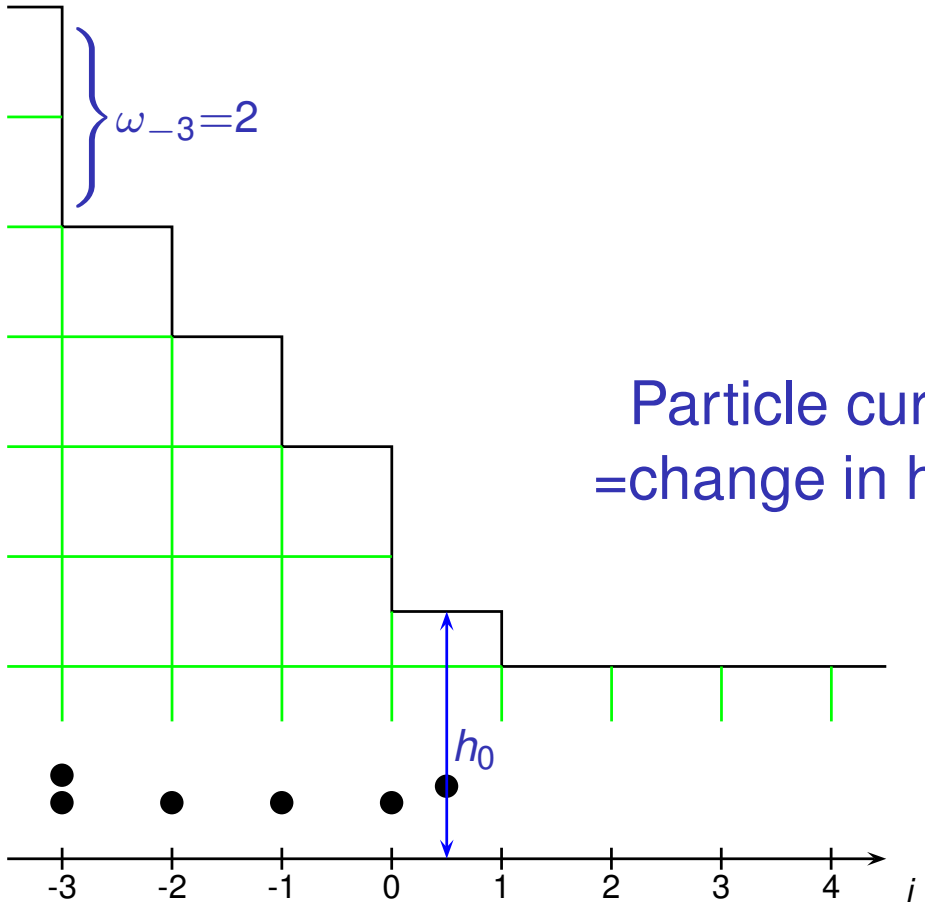
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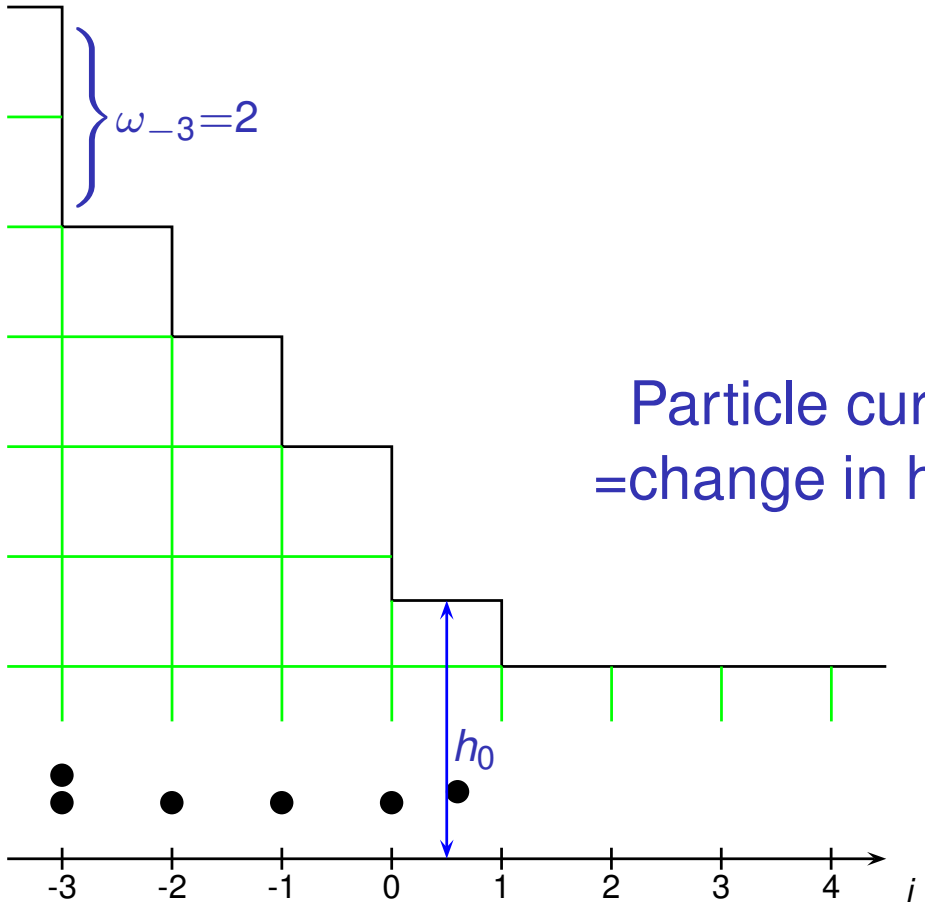
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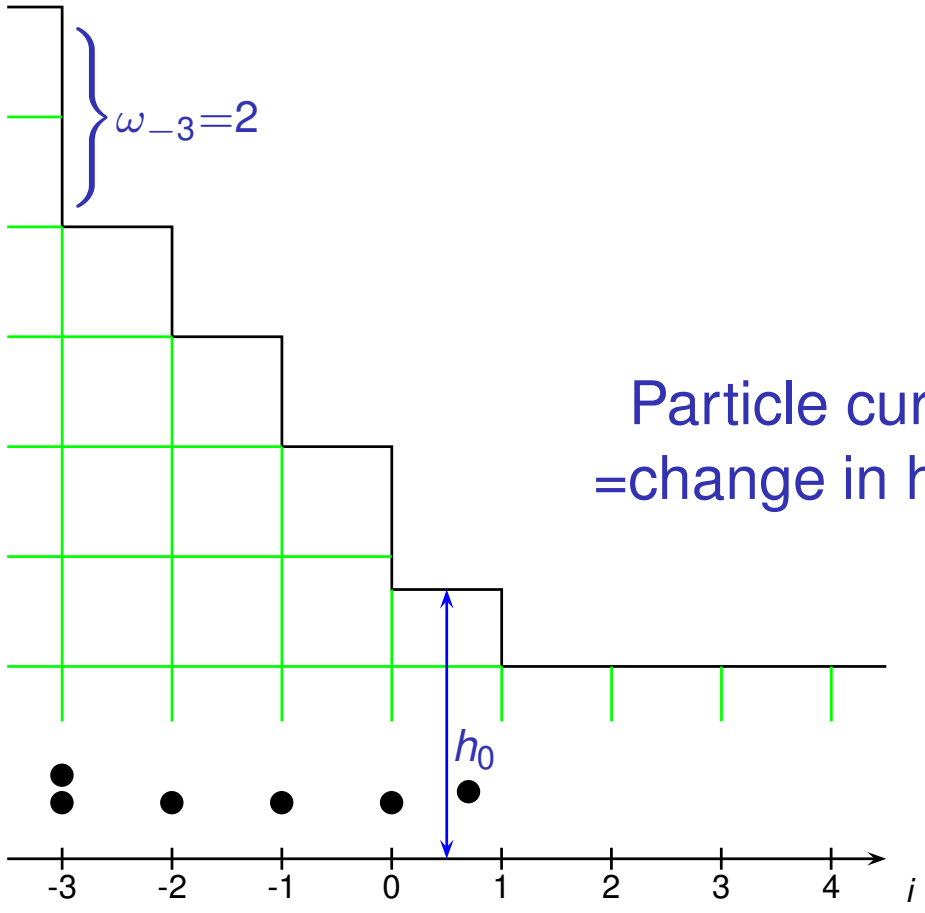
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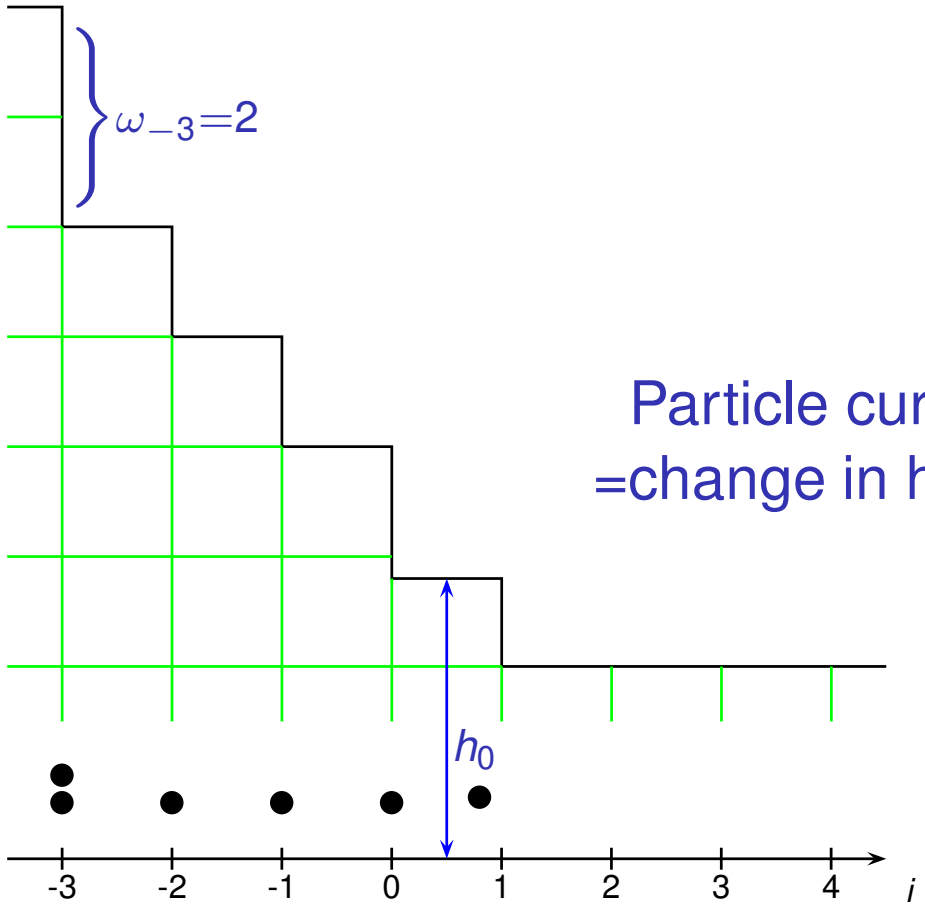
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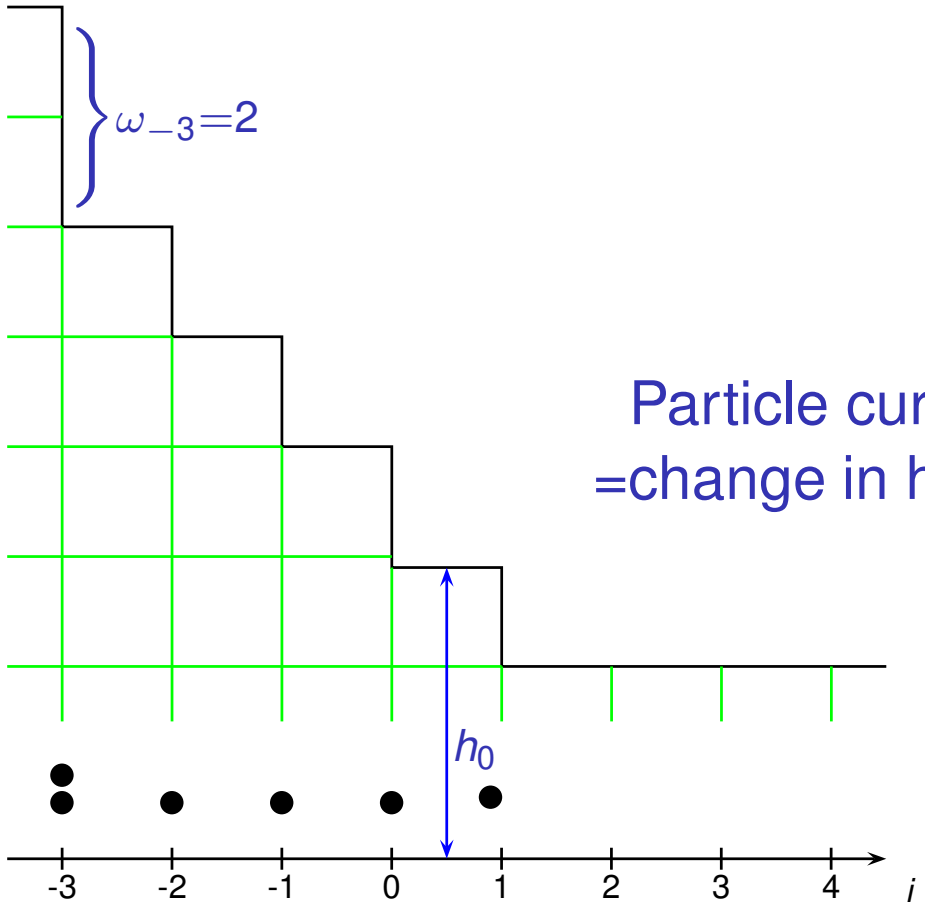
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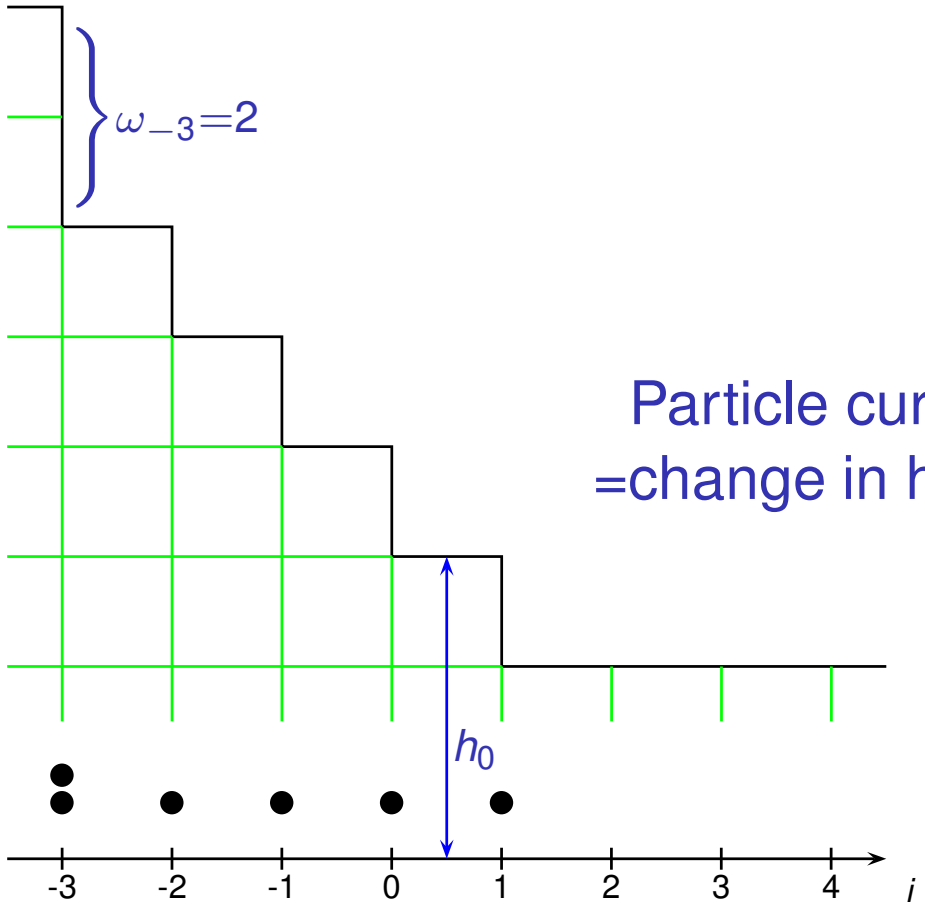
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to the right with rate $p \cdot f(\omega_i)$ (f non-decreasing)
to the left with rate $q \cdot f(\omega_i)$ ($q = 1 - p < p$).

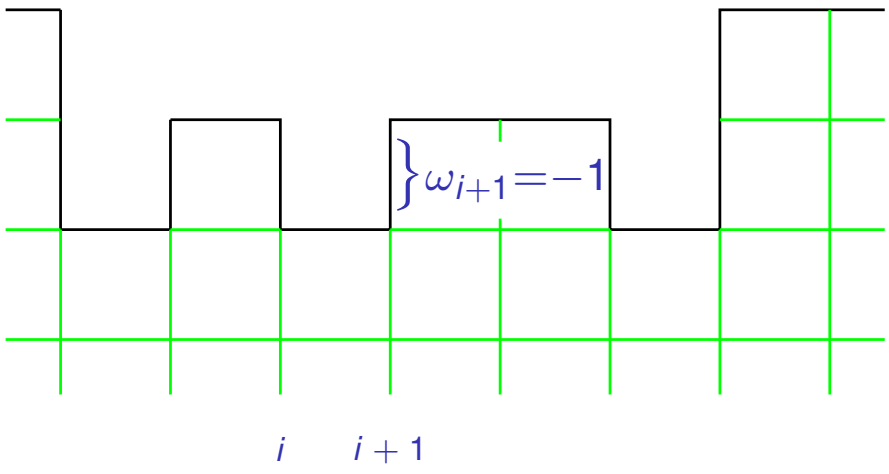
The asymmetric zero range process



Particle current
=change in height

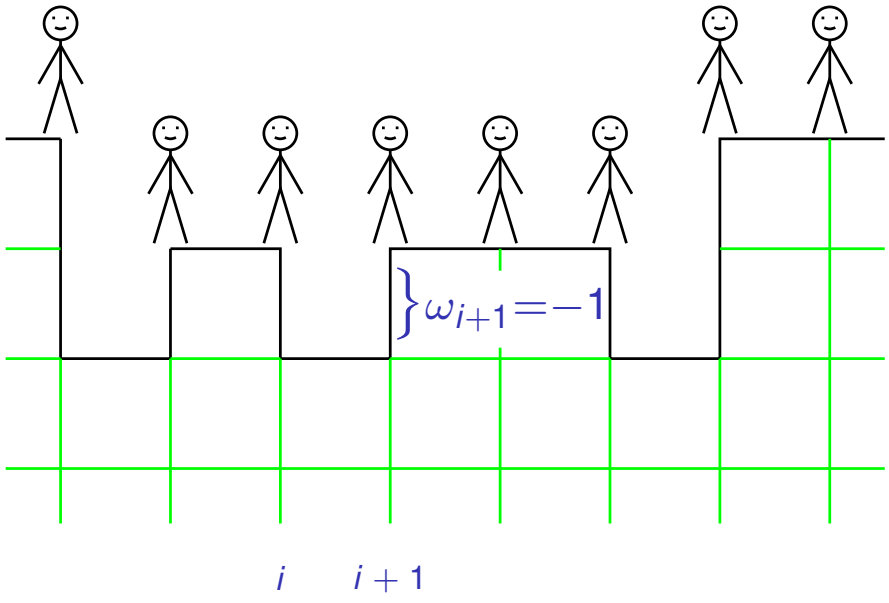
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The asymmetric bricklayers process (Bálint, 26 years ago)



$$\begin{pmatrix} \omega_i \\ \omega_{i+1} \end{pmatrix} \\ || \\ \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

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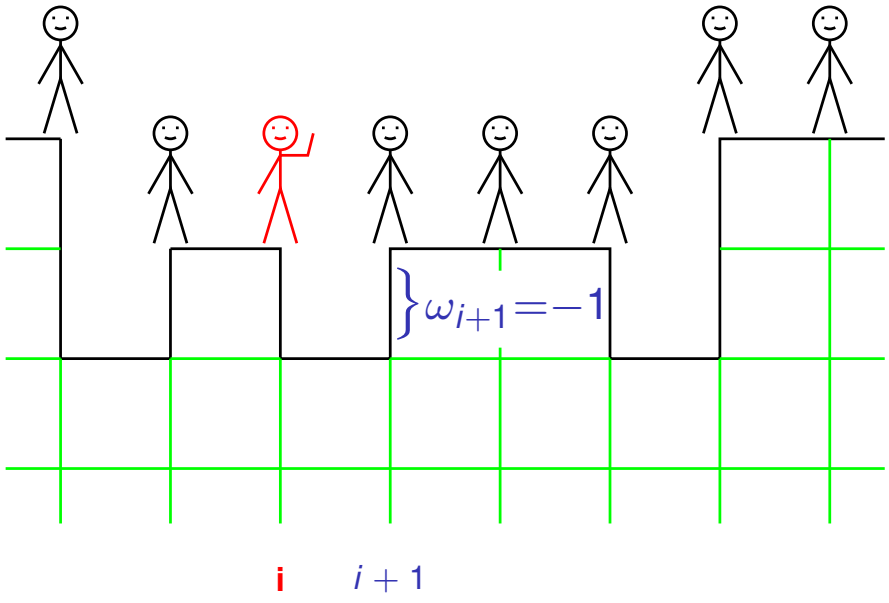


$$\begin{pmatrix} \omega_i \\ \omega_{i+1} \end{pmatrix} \quad || \quad \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

a brick is added with rate $p \cdot [f(\omega_i) + f(-\omega_{i+1})]$
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$(f(\omega) \cdot f(1-\omega) = 1; \quad f$ non-decreasing; $\quad q = 1 - p < p)$.

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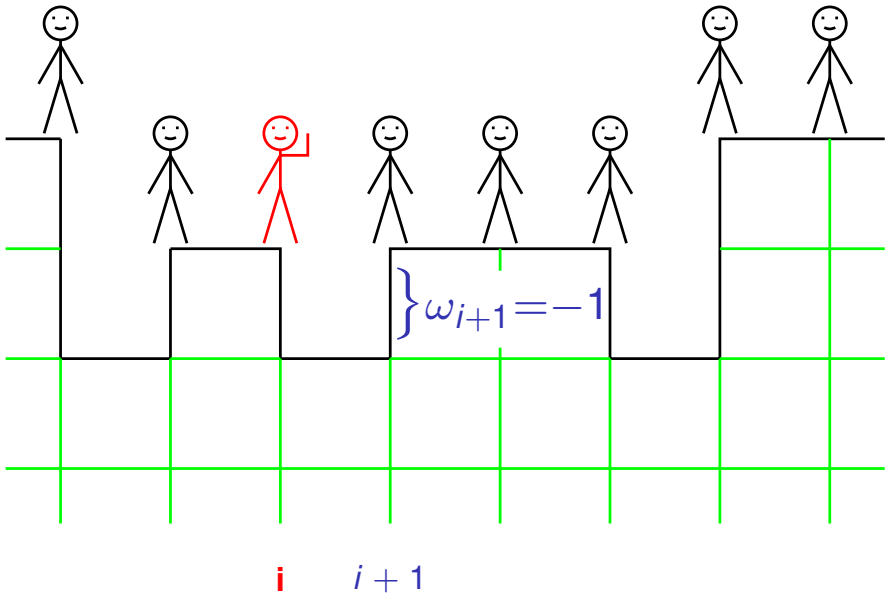


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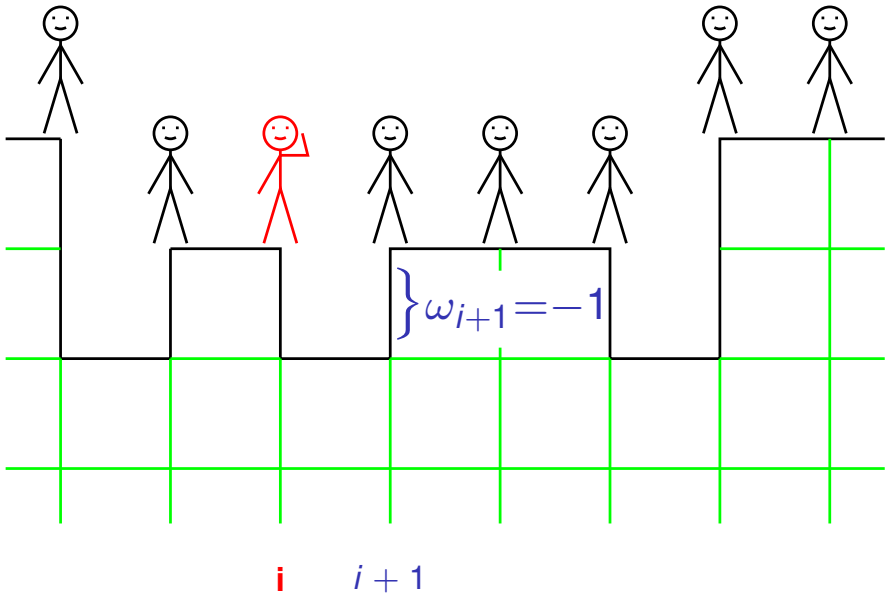


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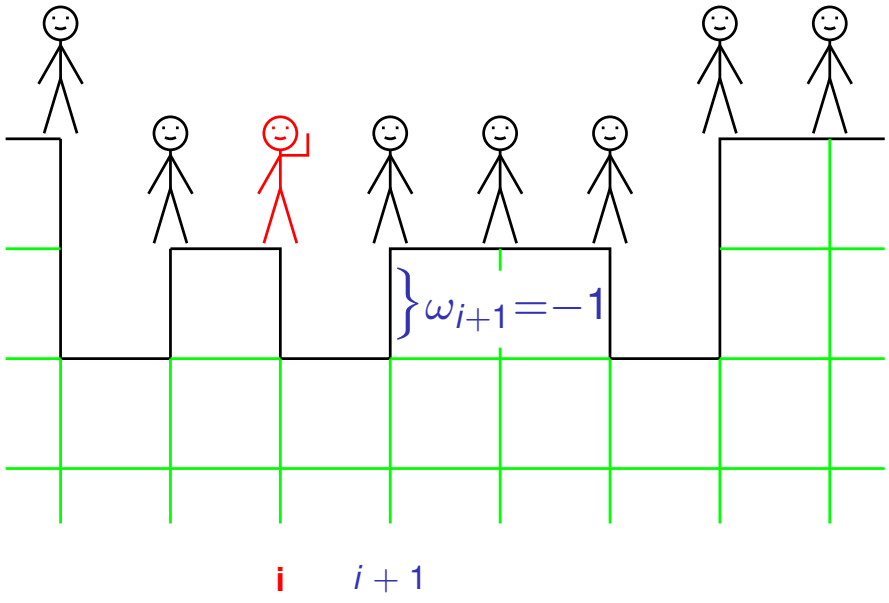


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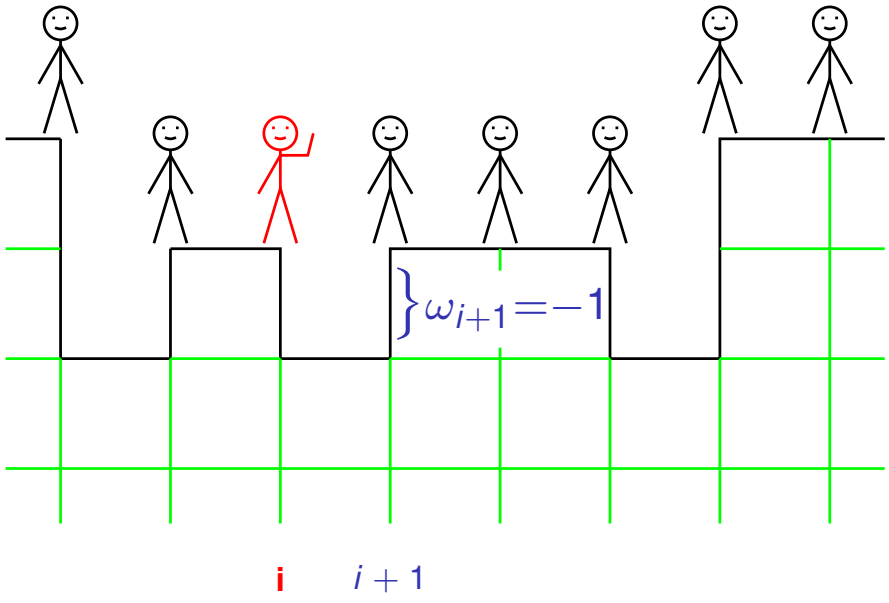


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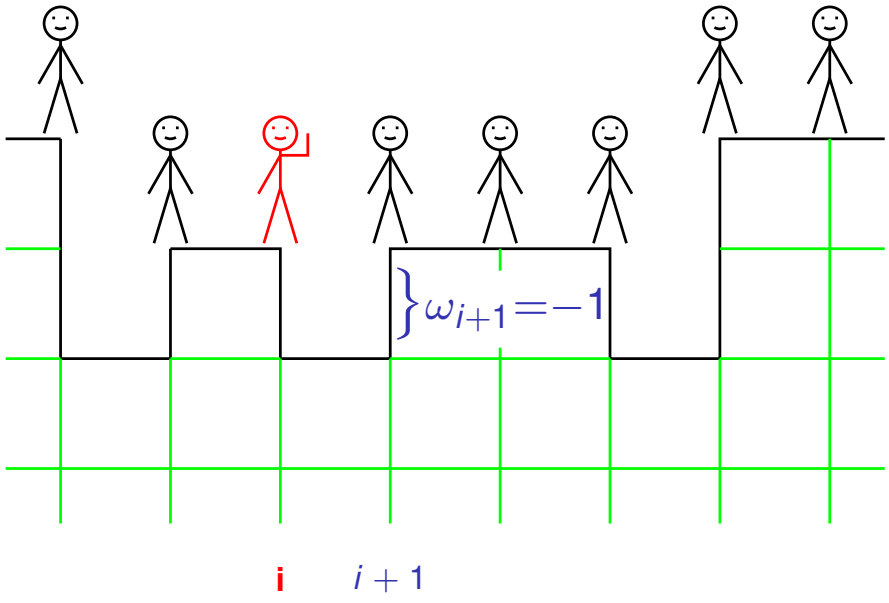


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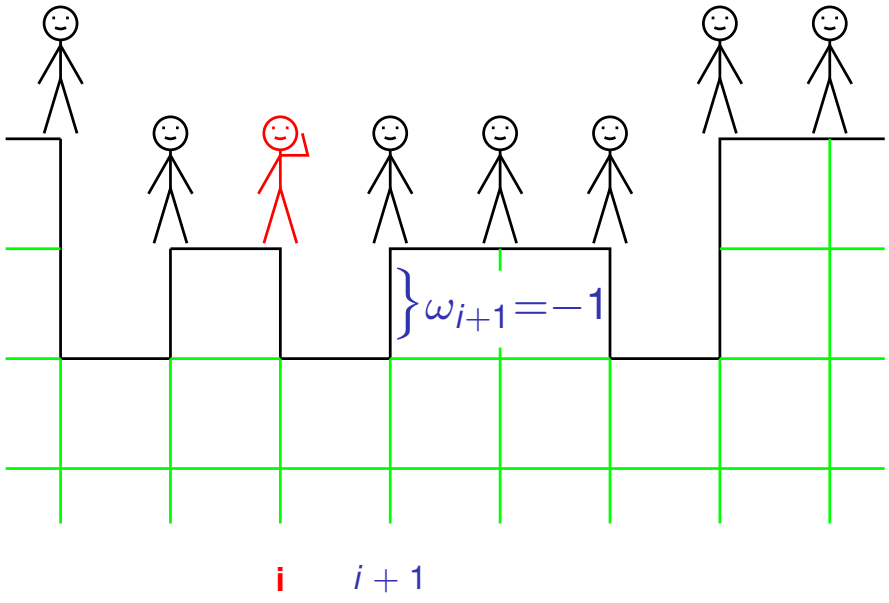


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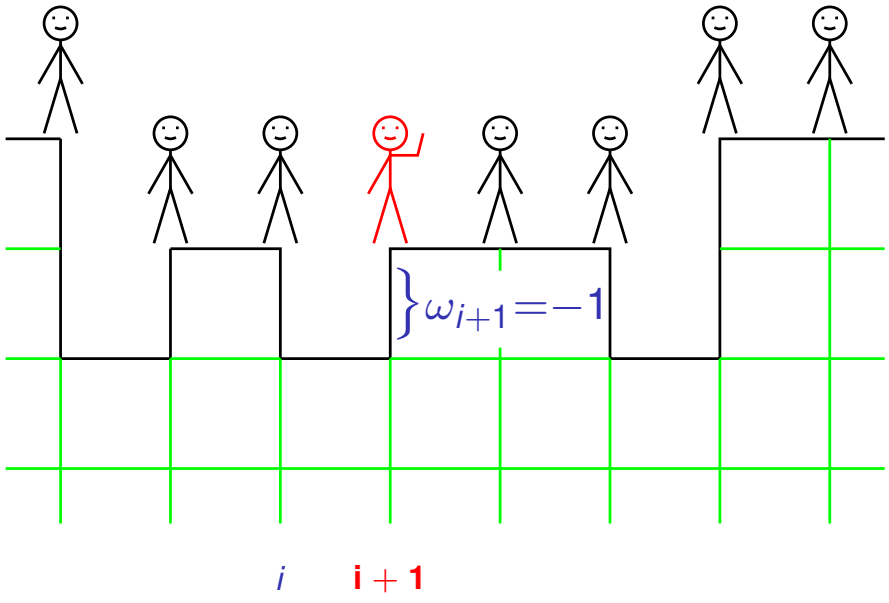


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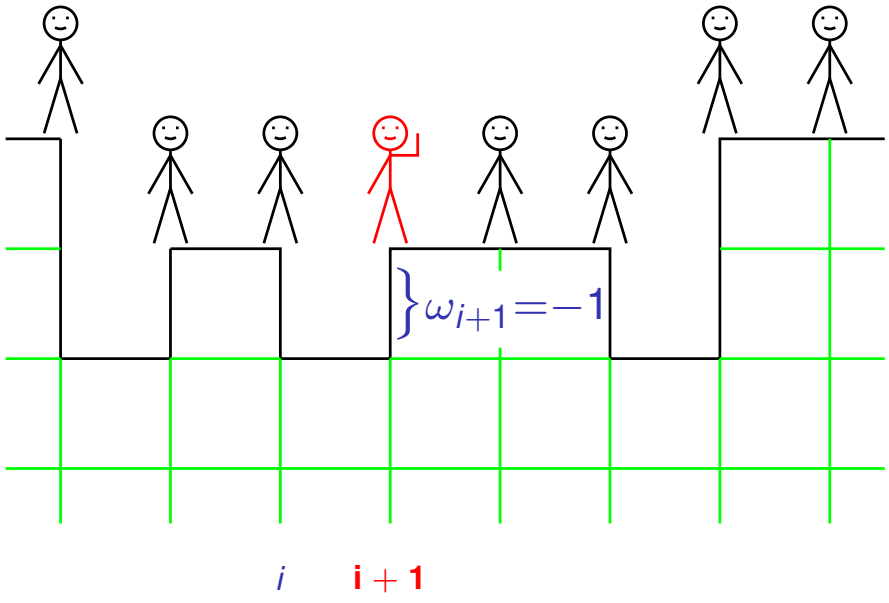


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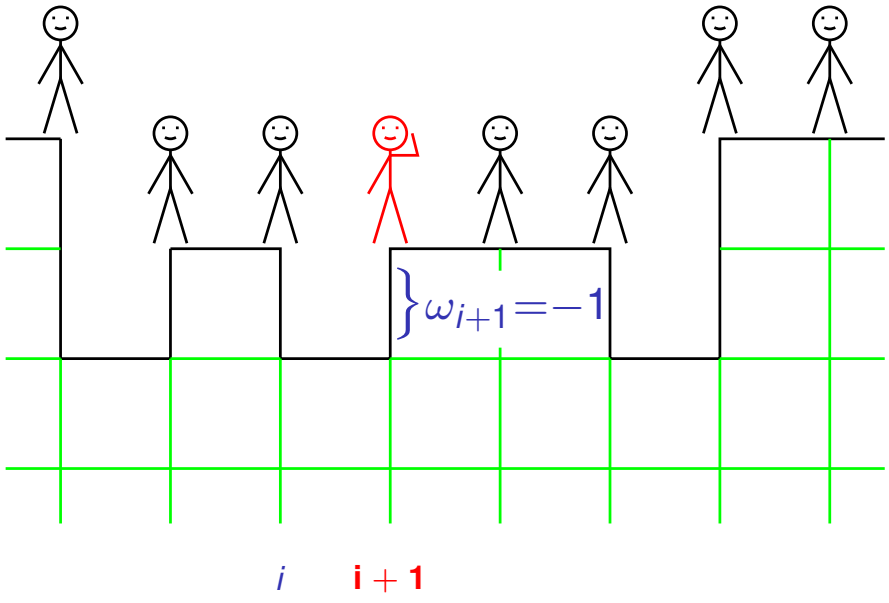


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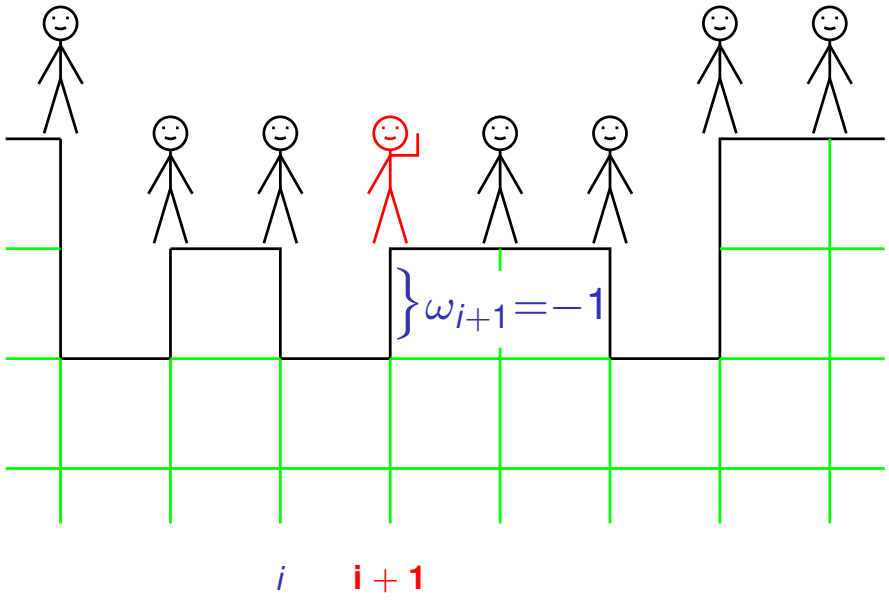


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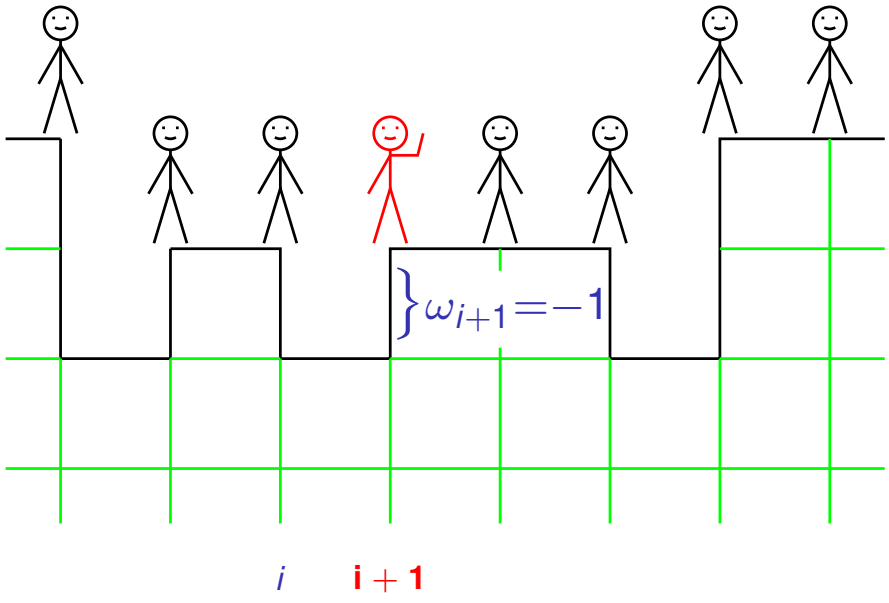


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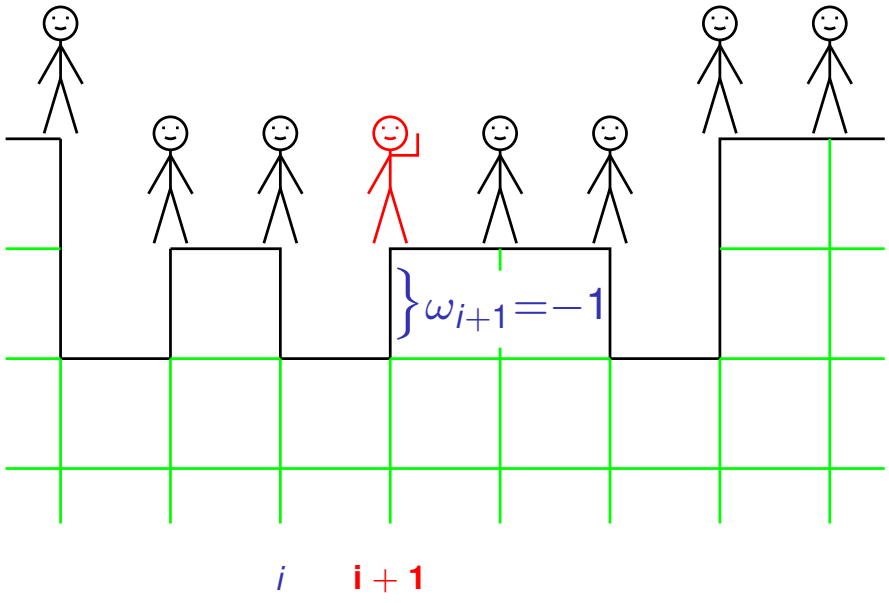


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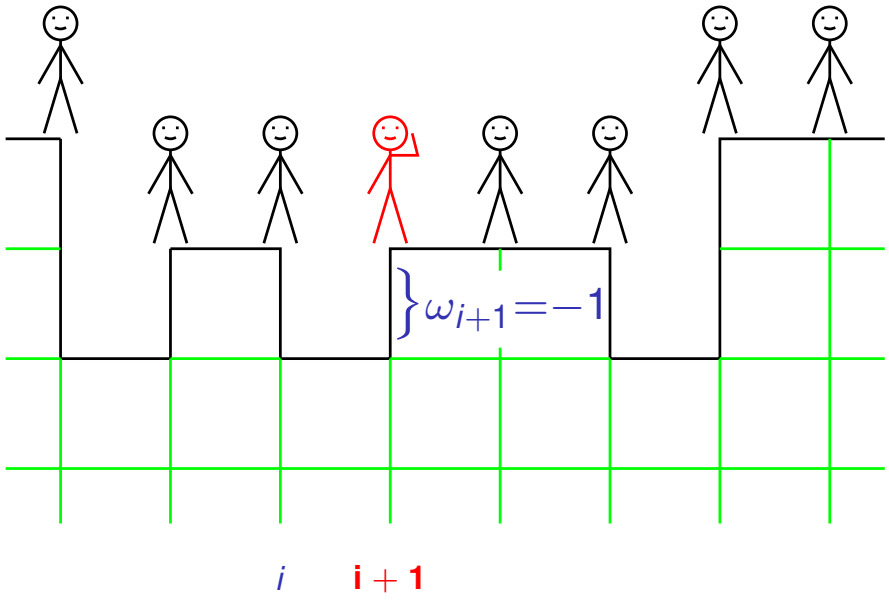


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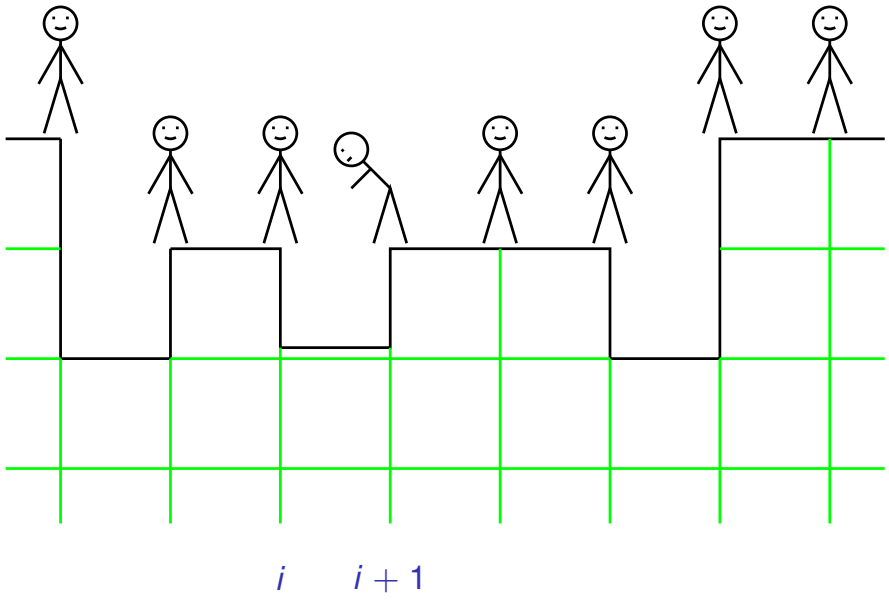


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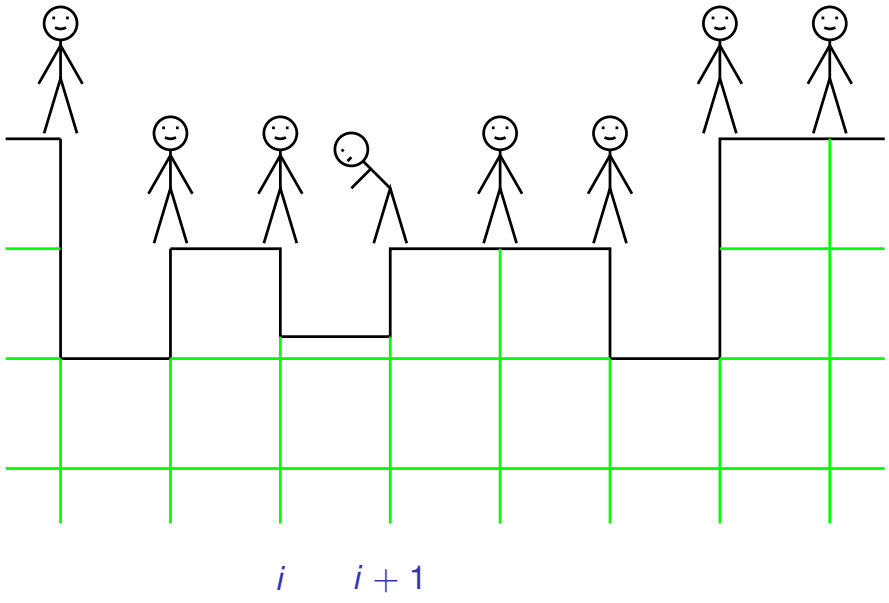


$$\begin{pmatrix} \omega_i \\ \omega_{i+1} \end{pmatrix} \to \begin{pmatrix} \omega_i - 1 \\ \omega_{i+1} + 1 \end{pmatrix}$$
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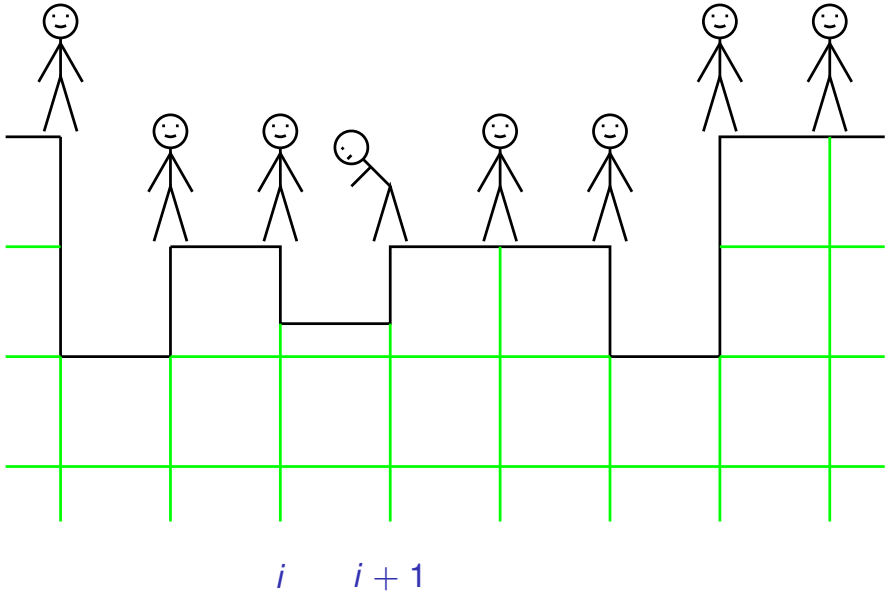


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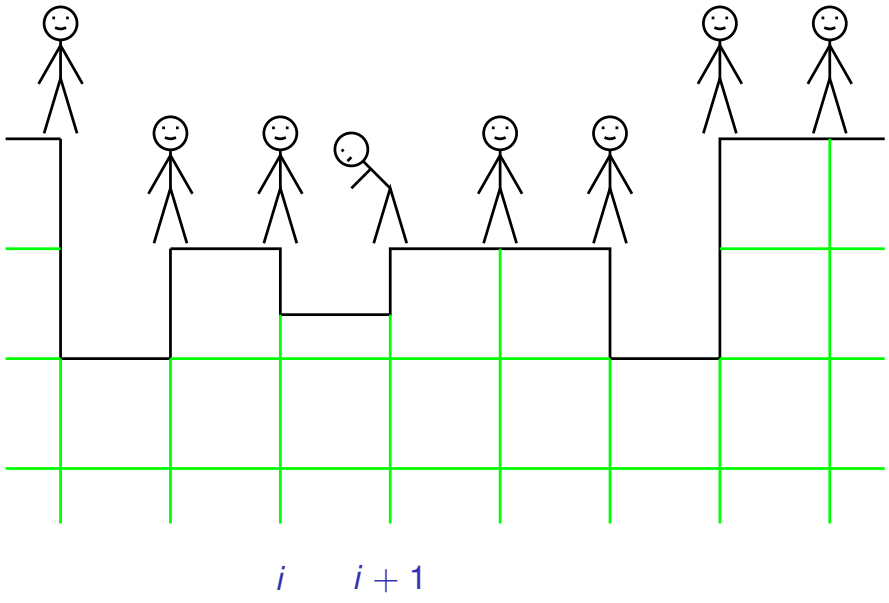


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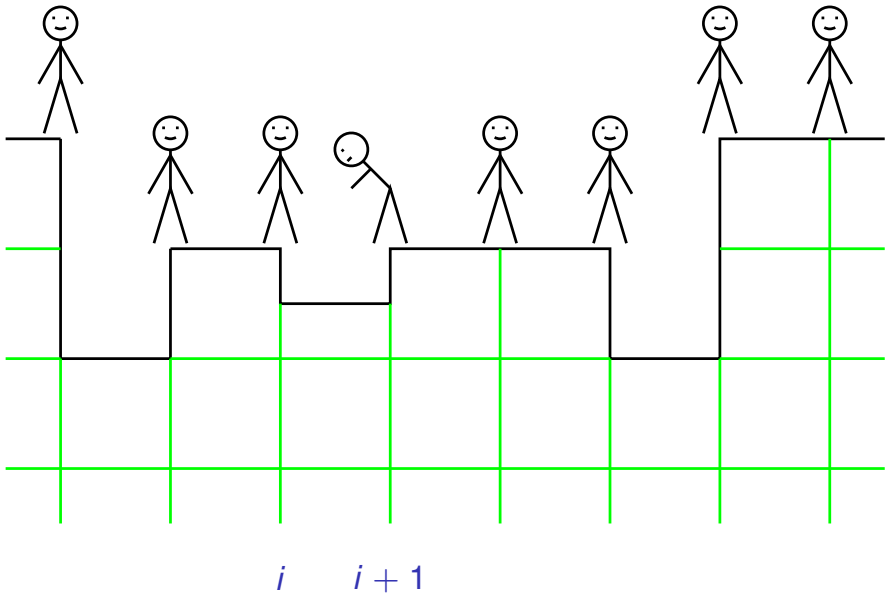


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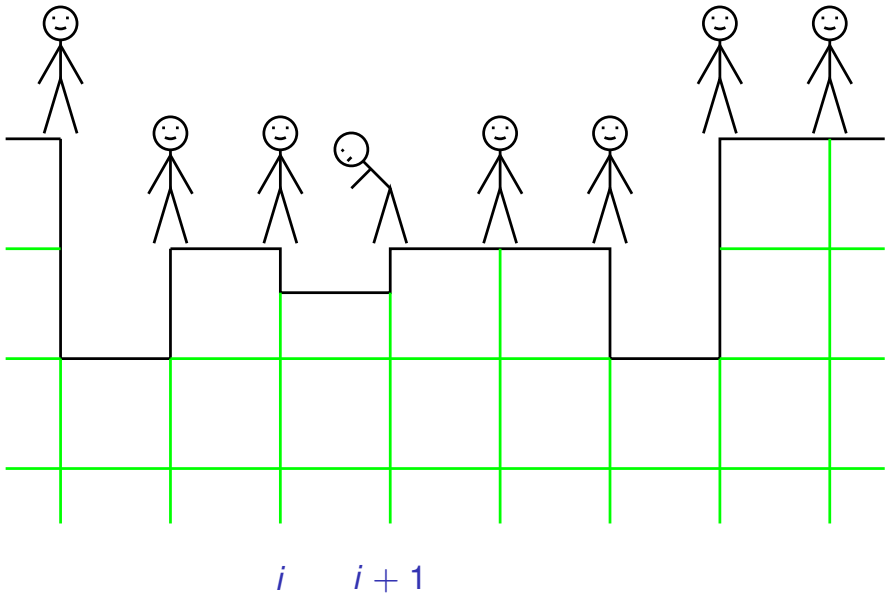


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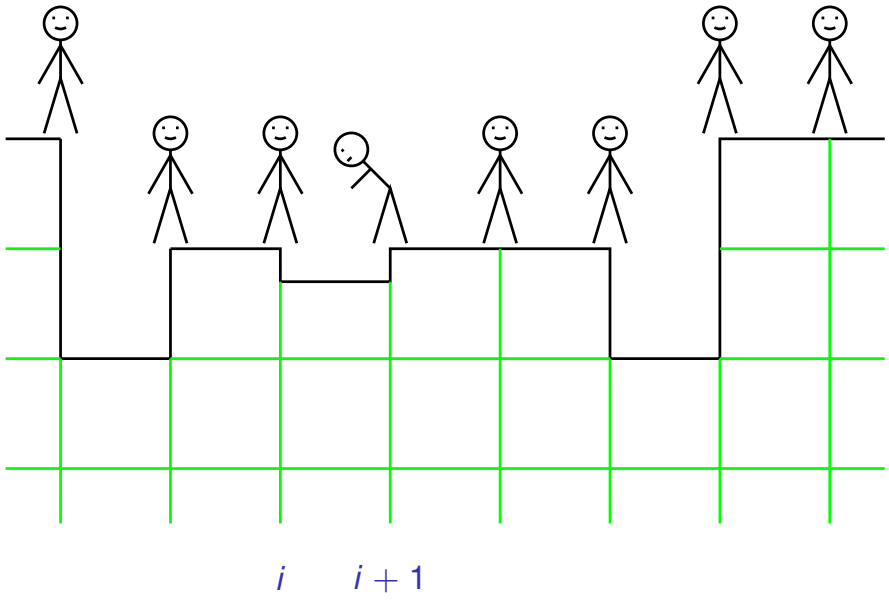


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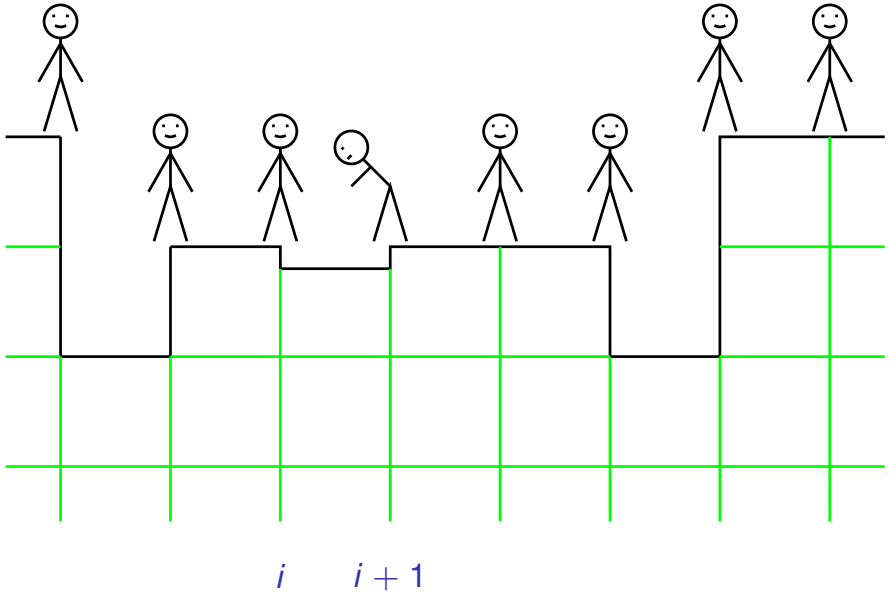


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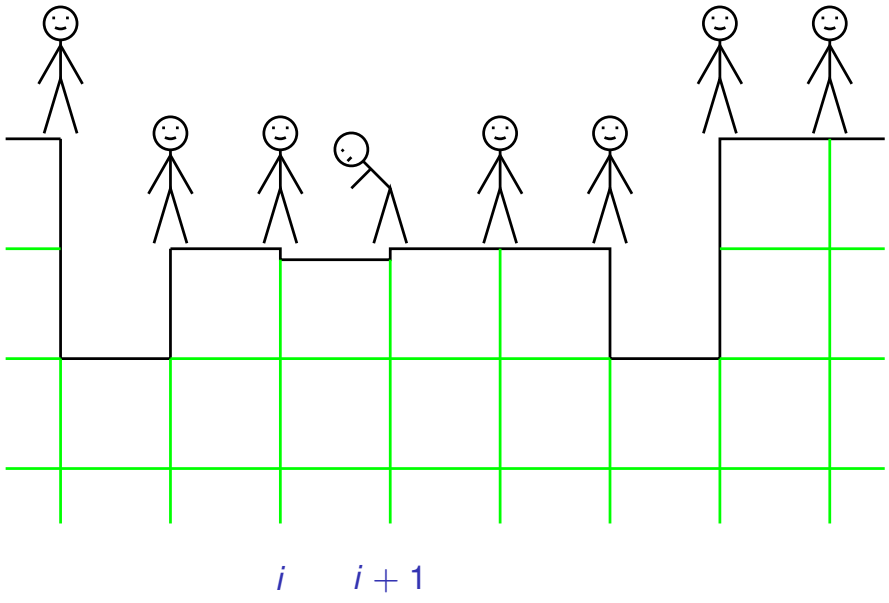


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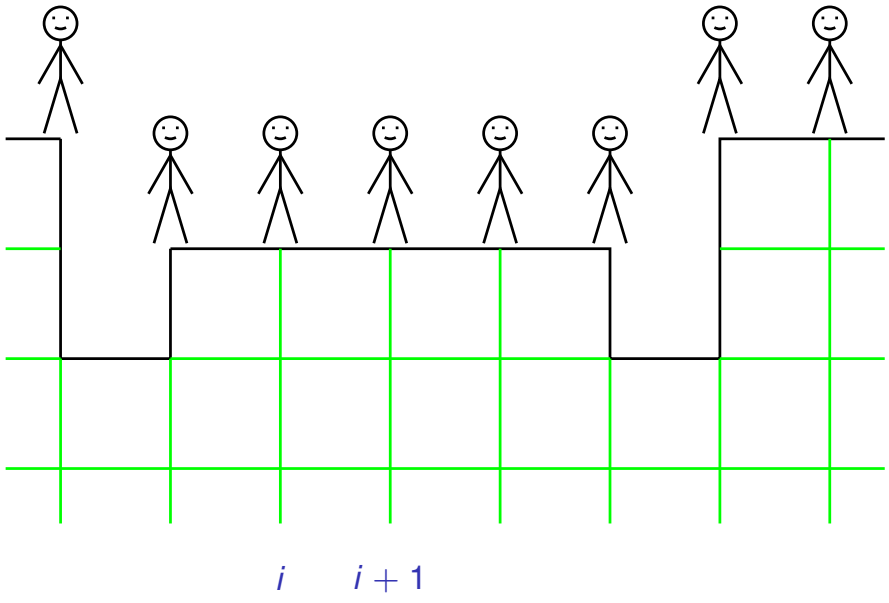


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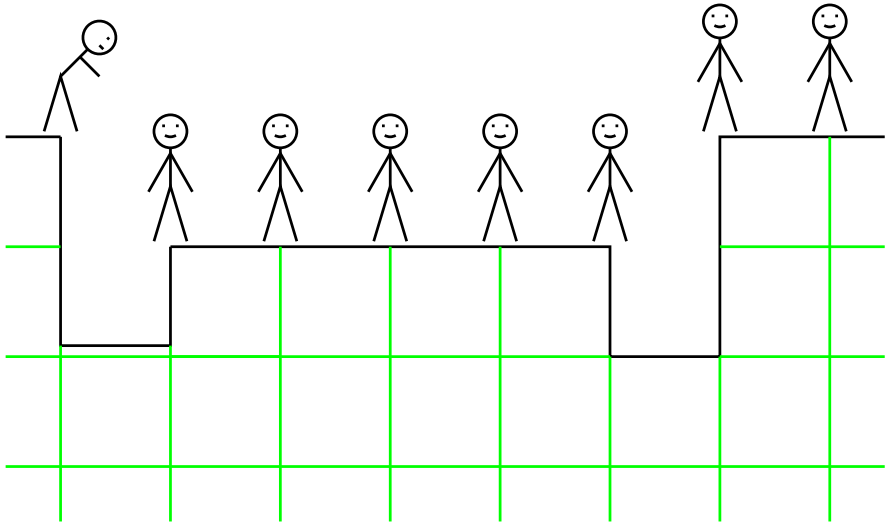


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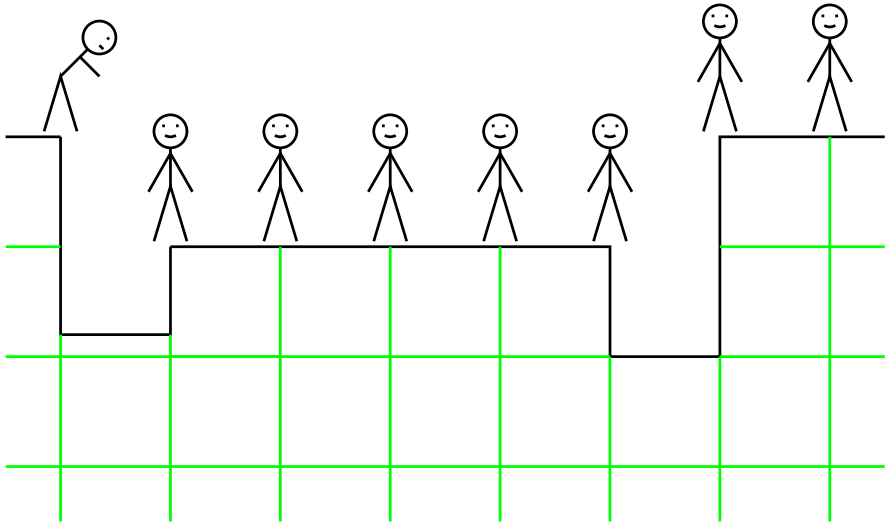
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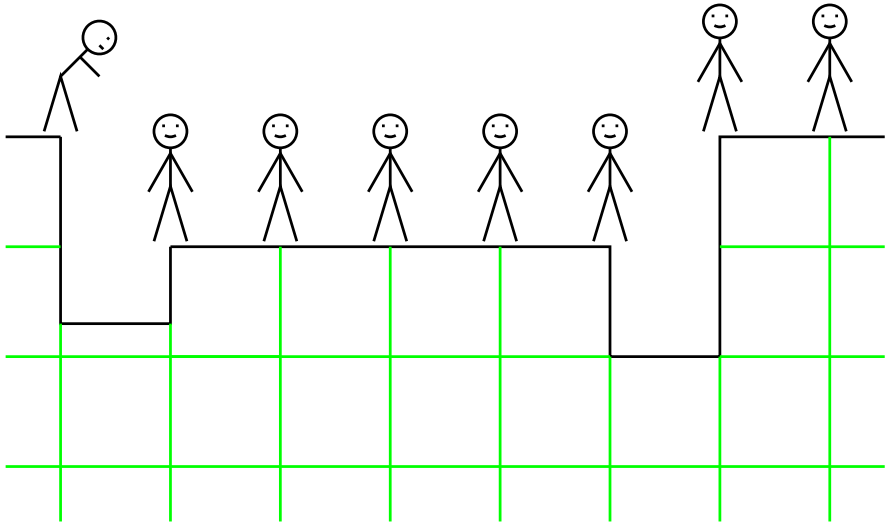
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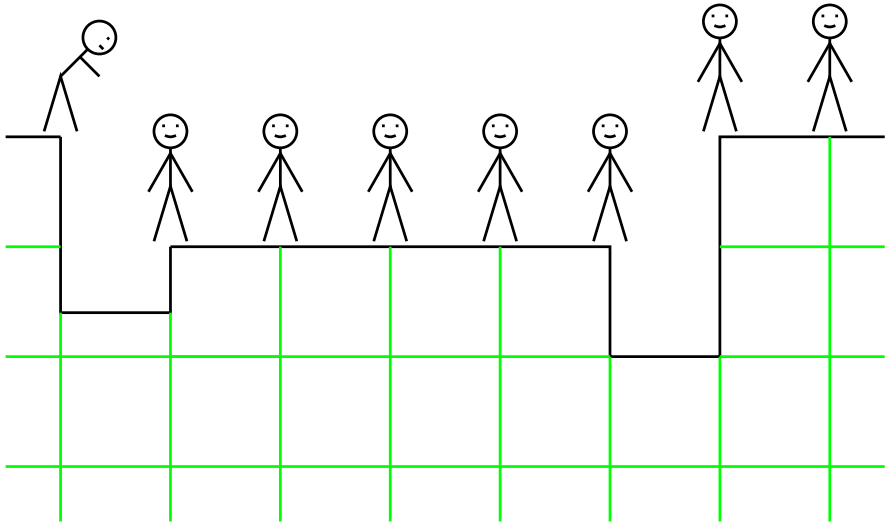
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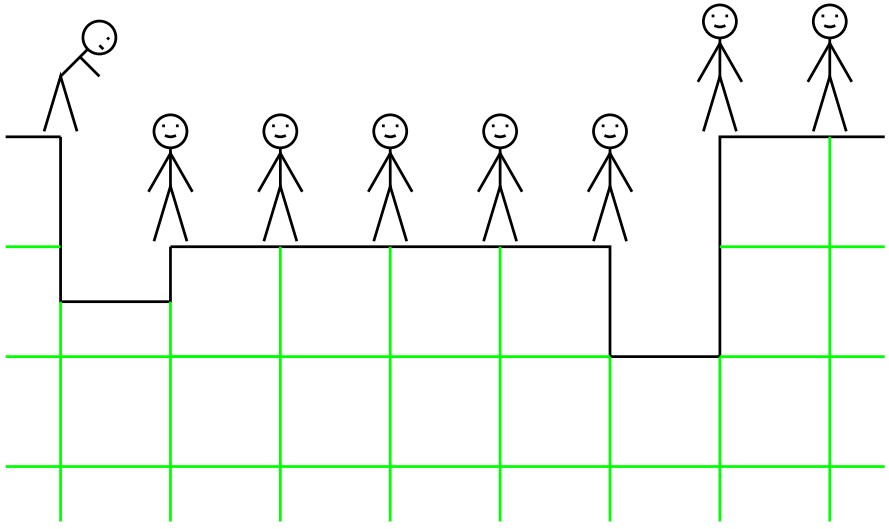
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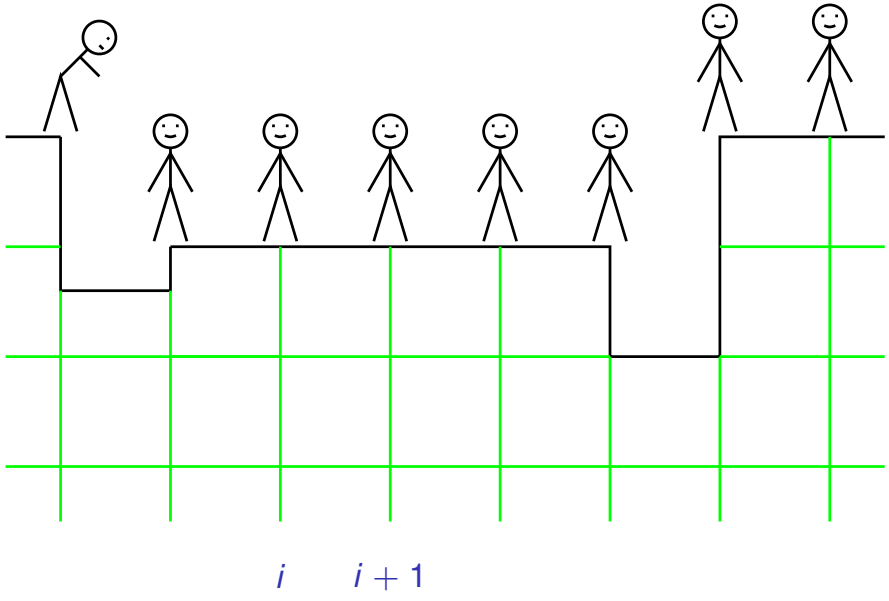
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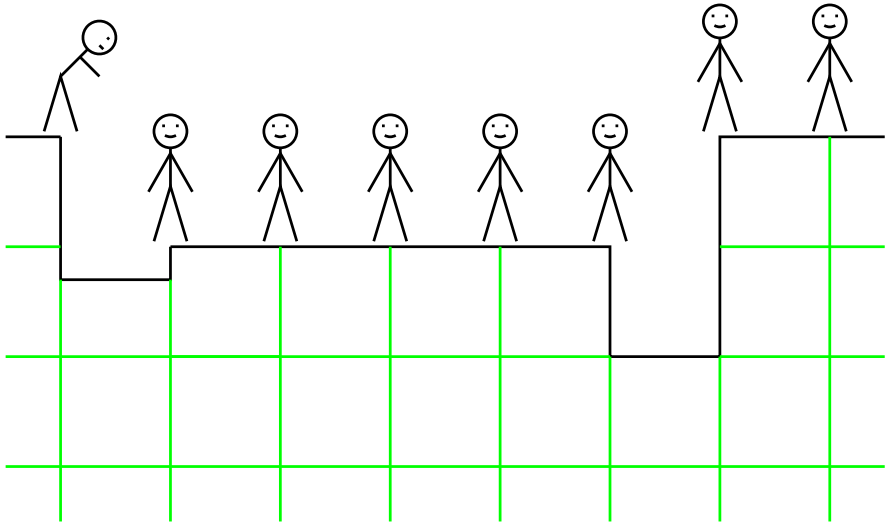
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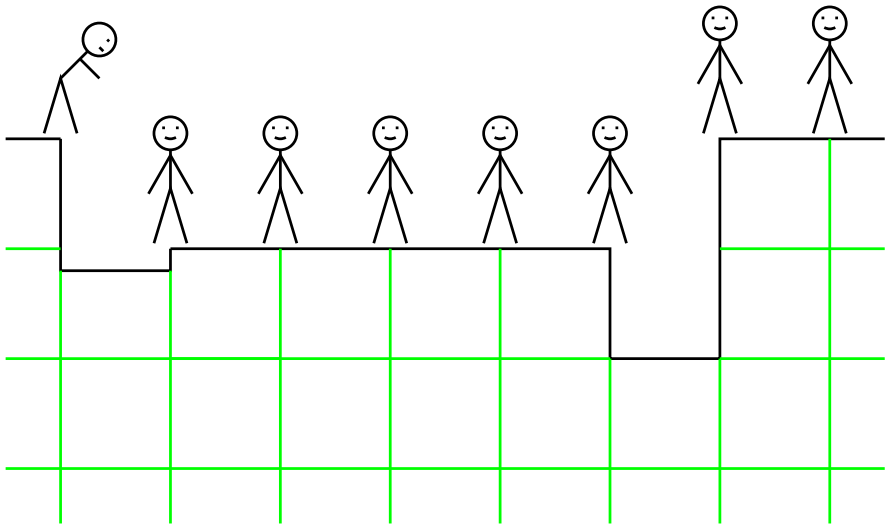
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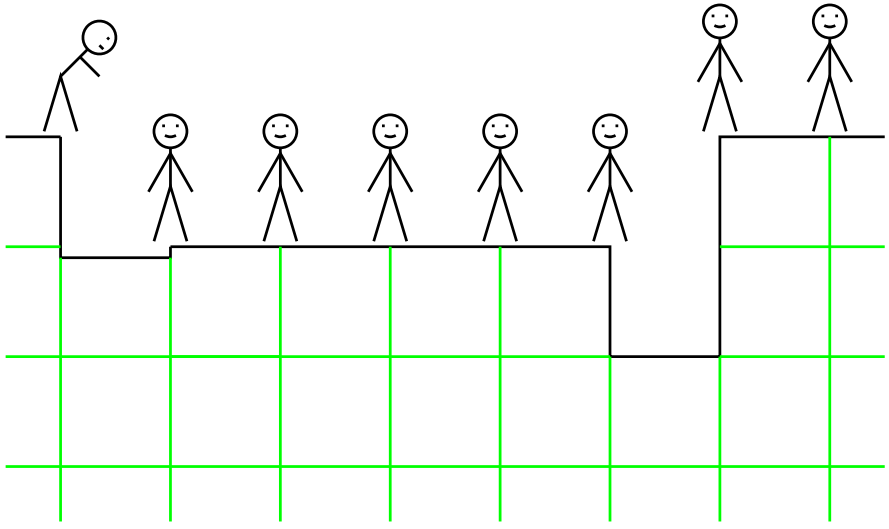
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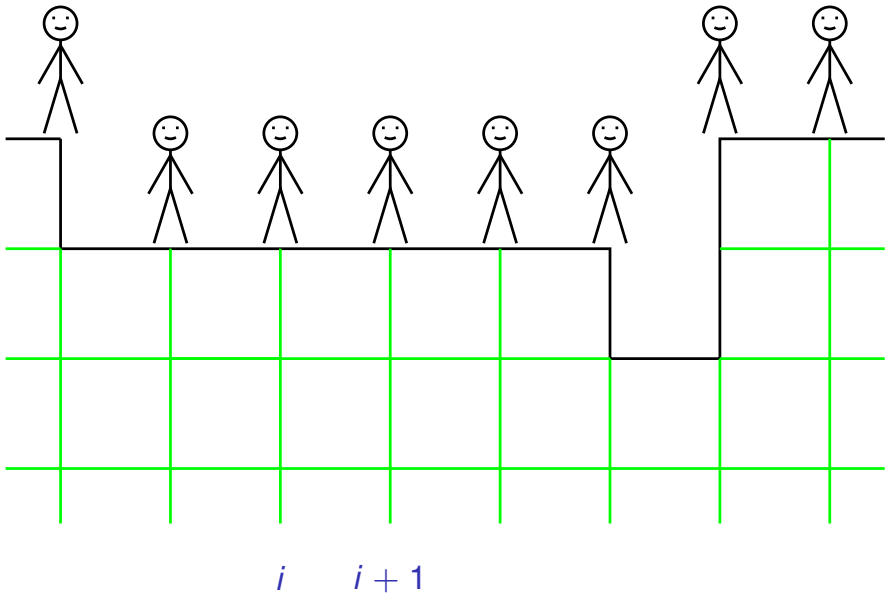
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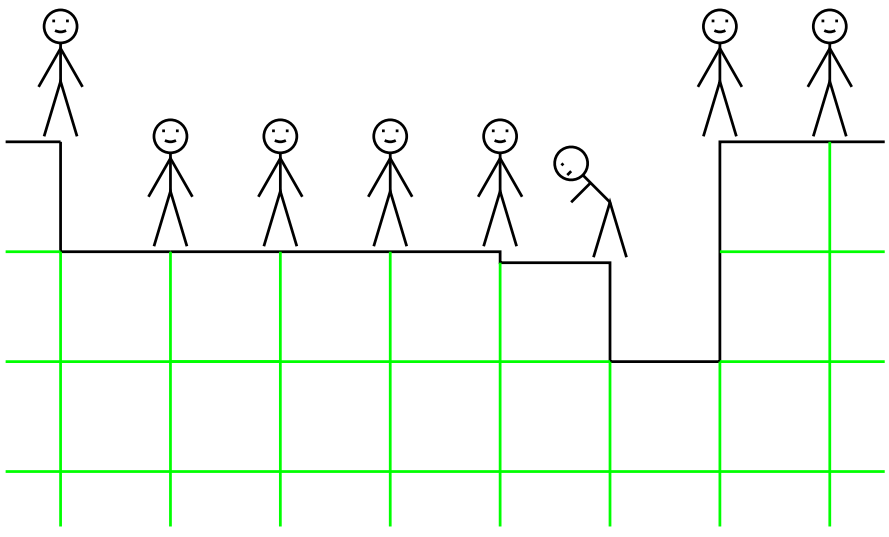
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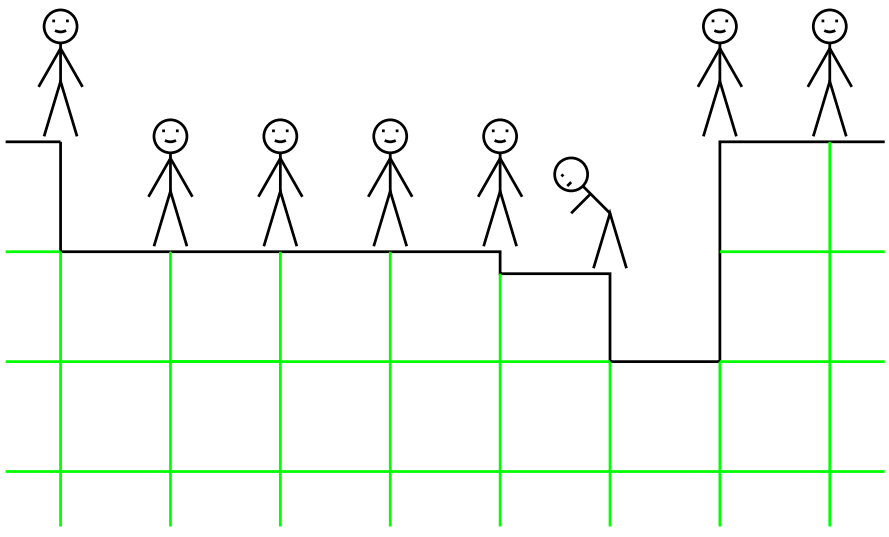
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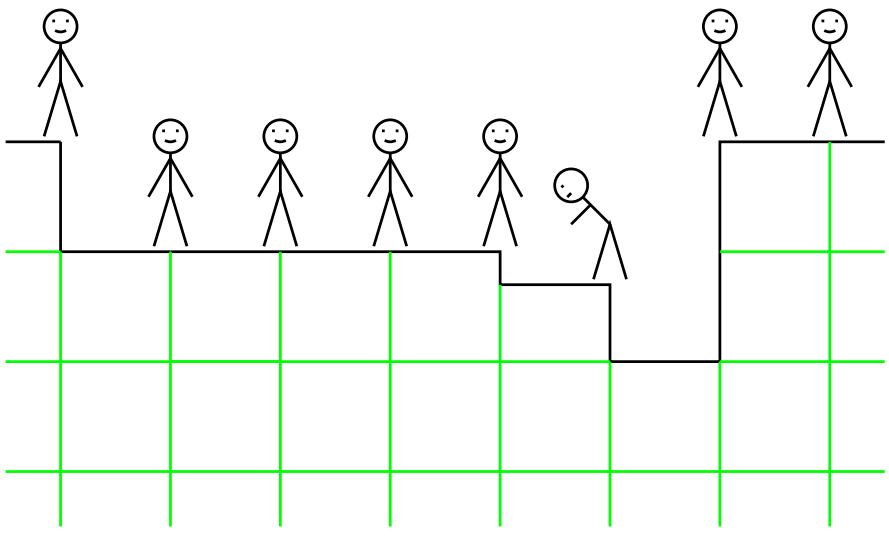
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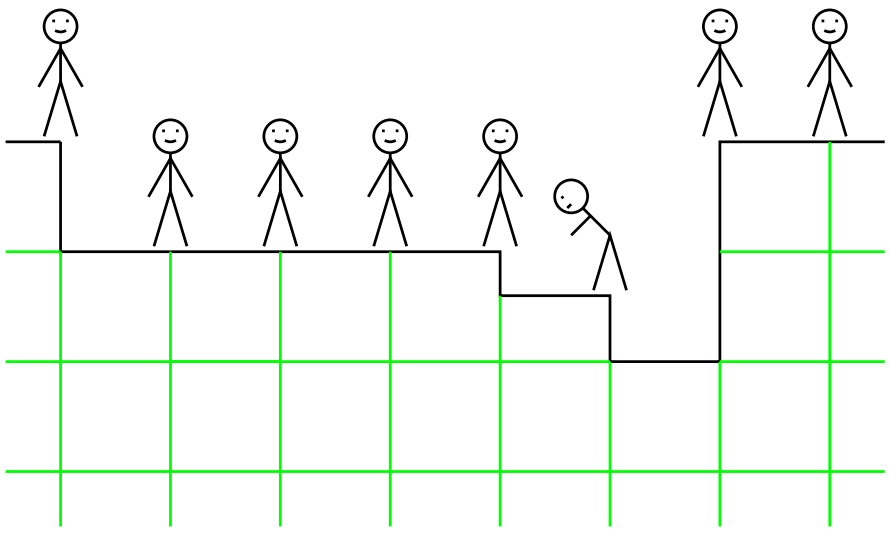
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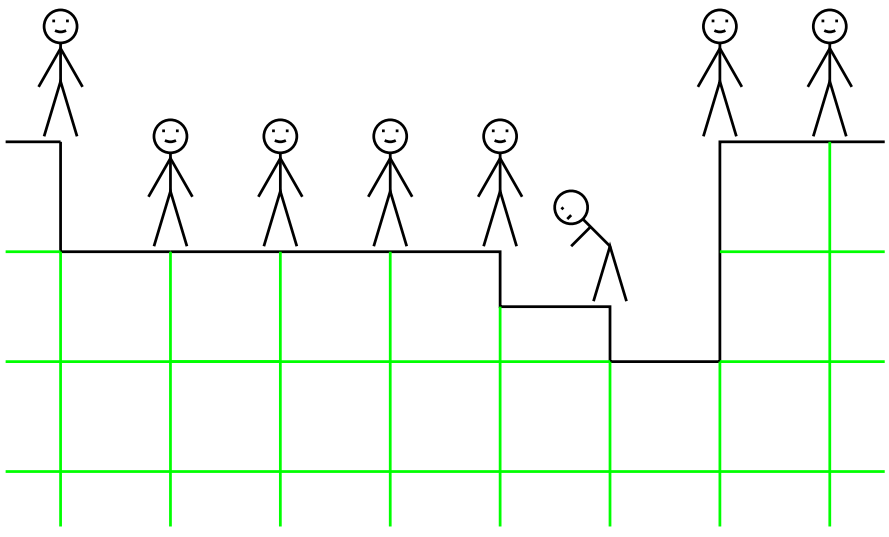
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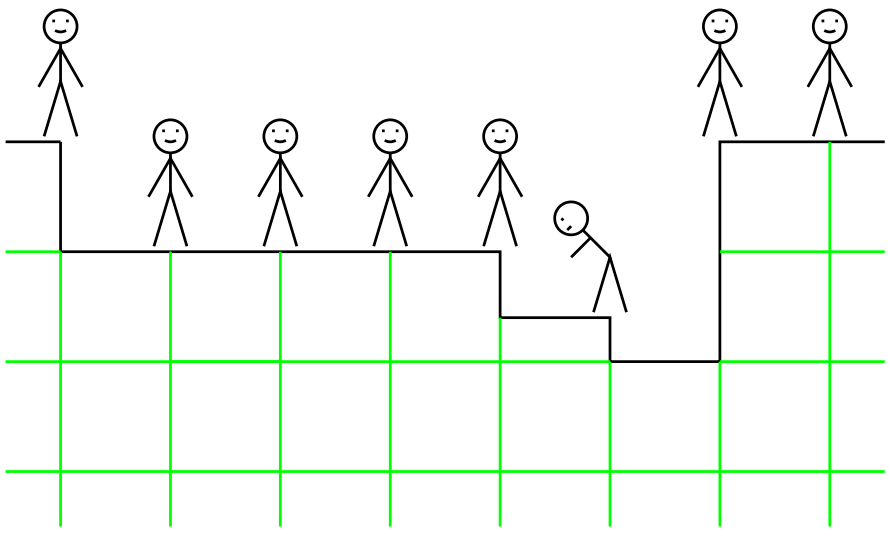
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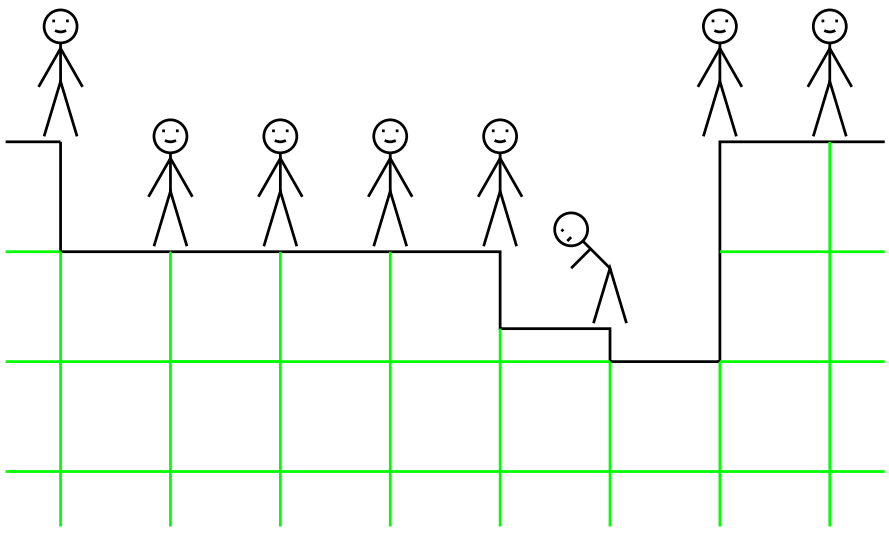
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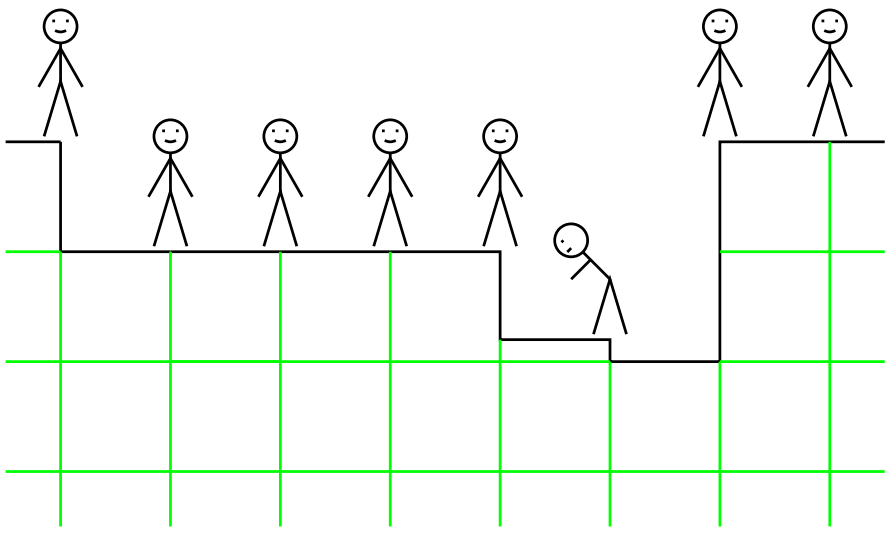
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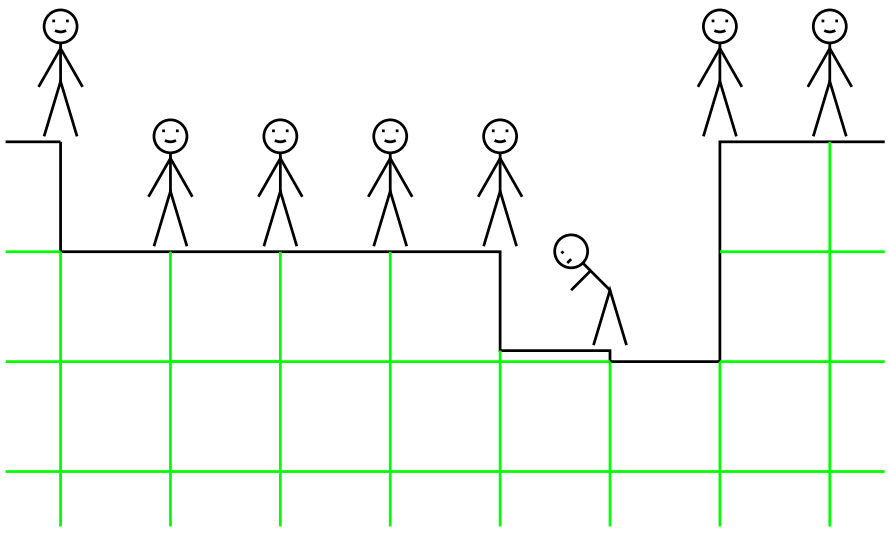
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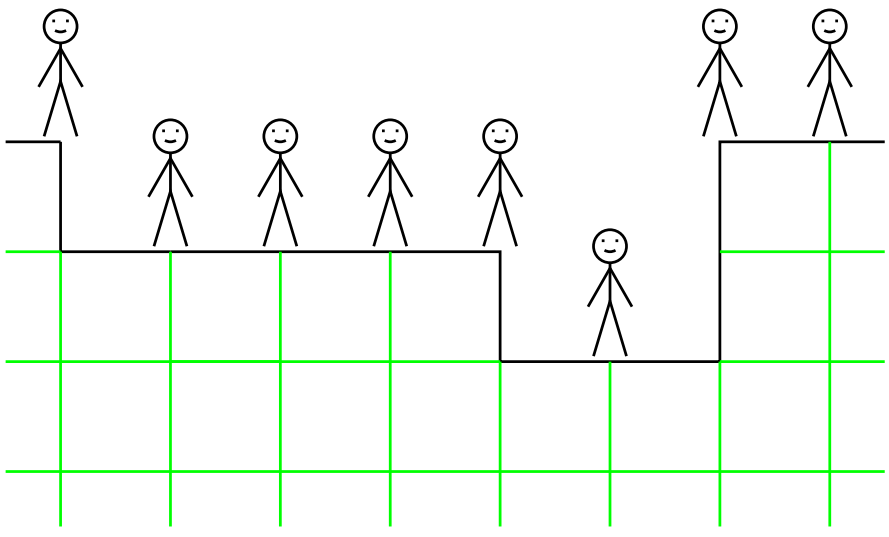
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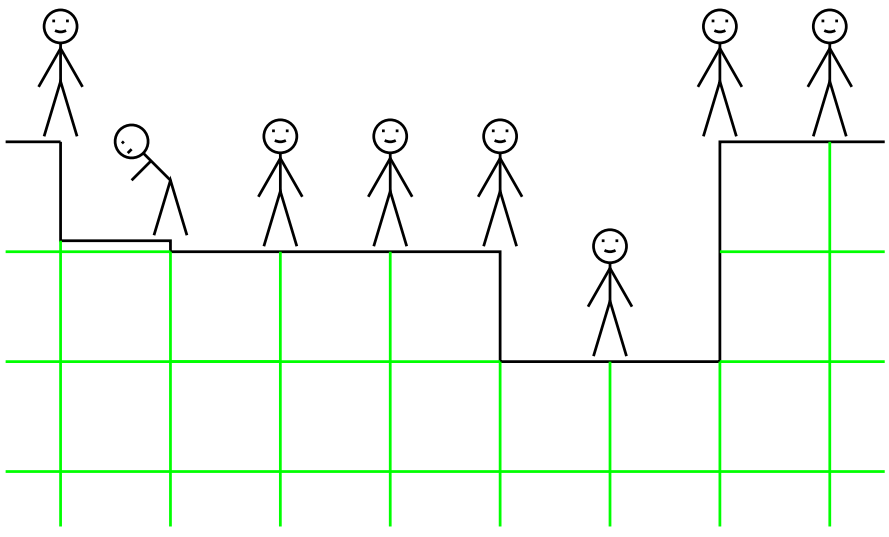
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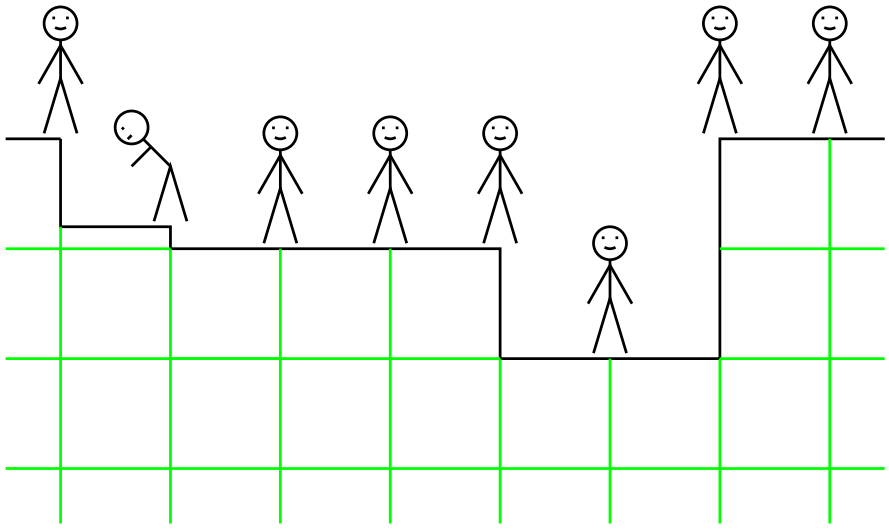
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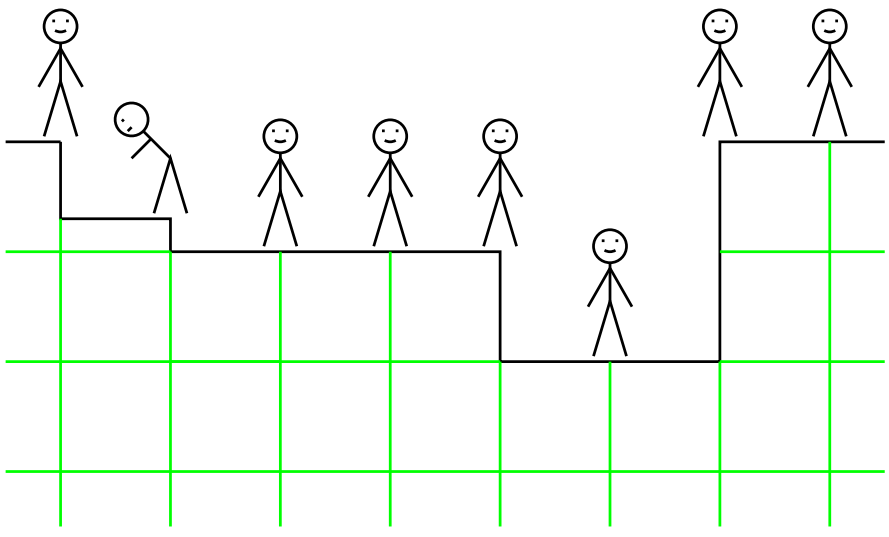
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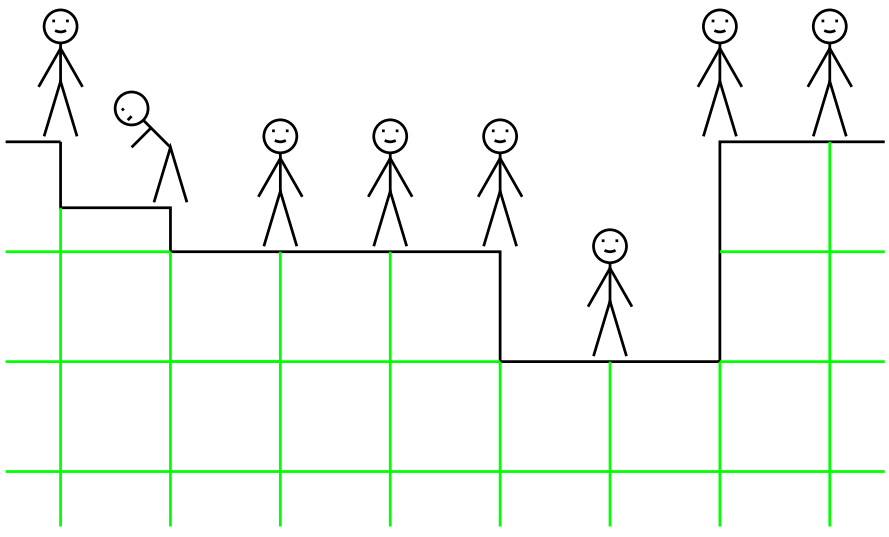
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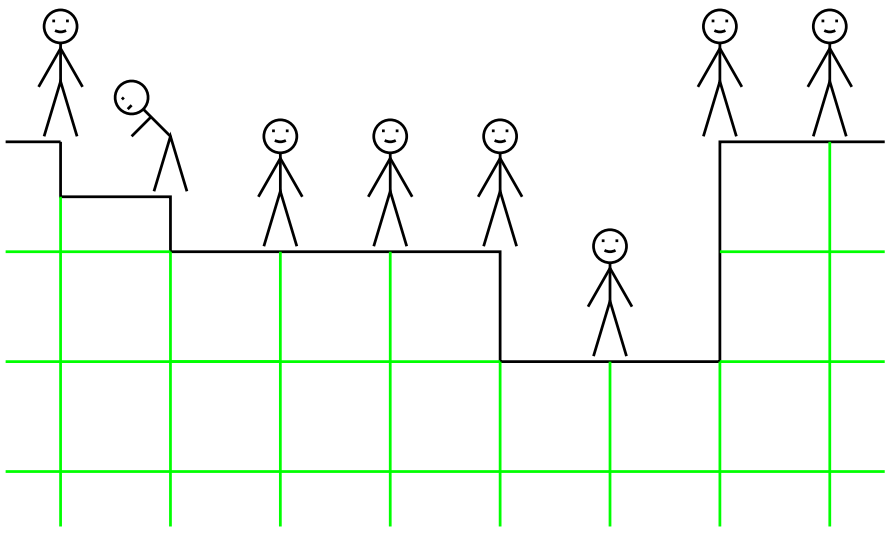
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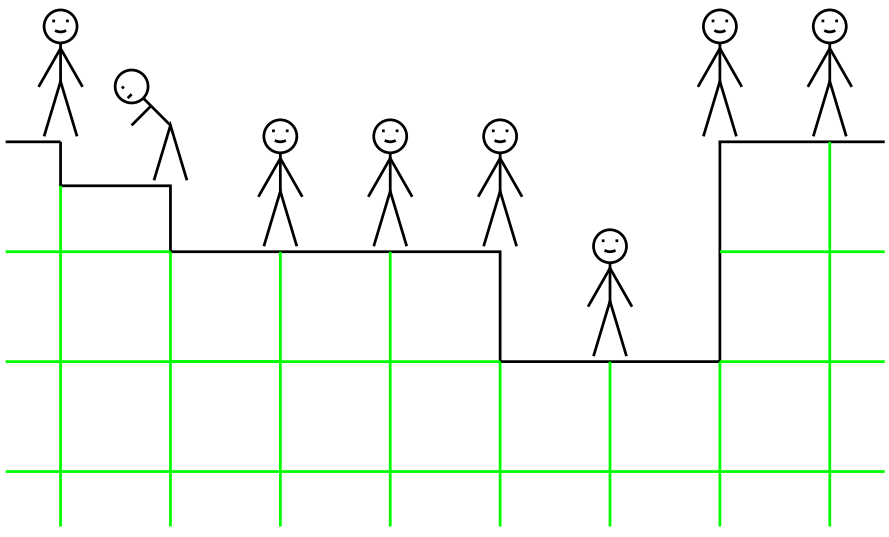
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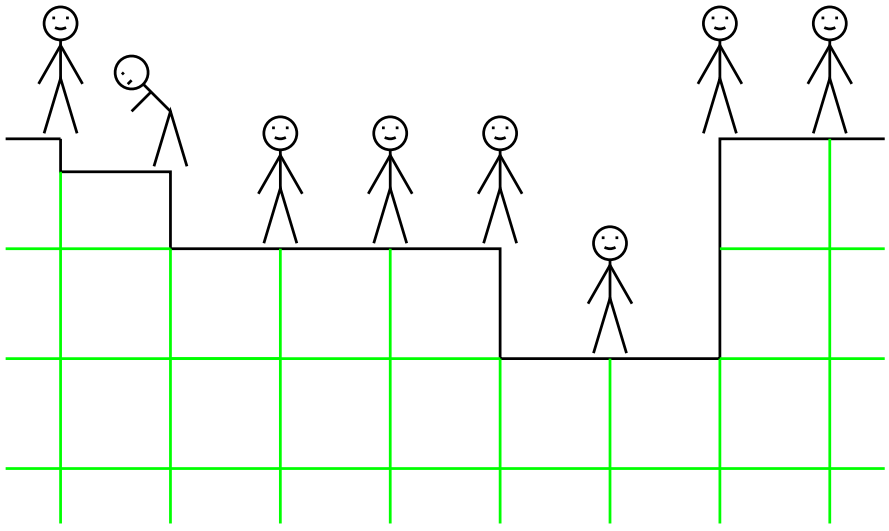
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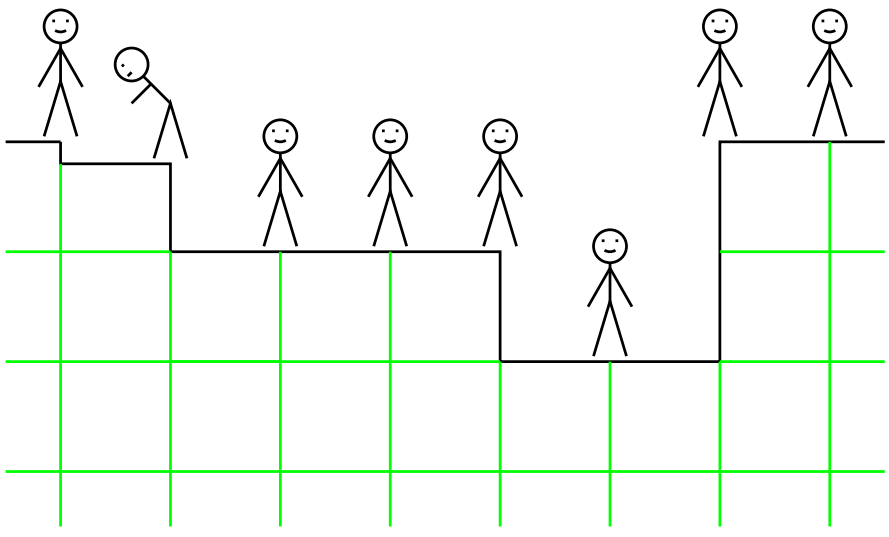
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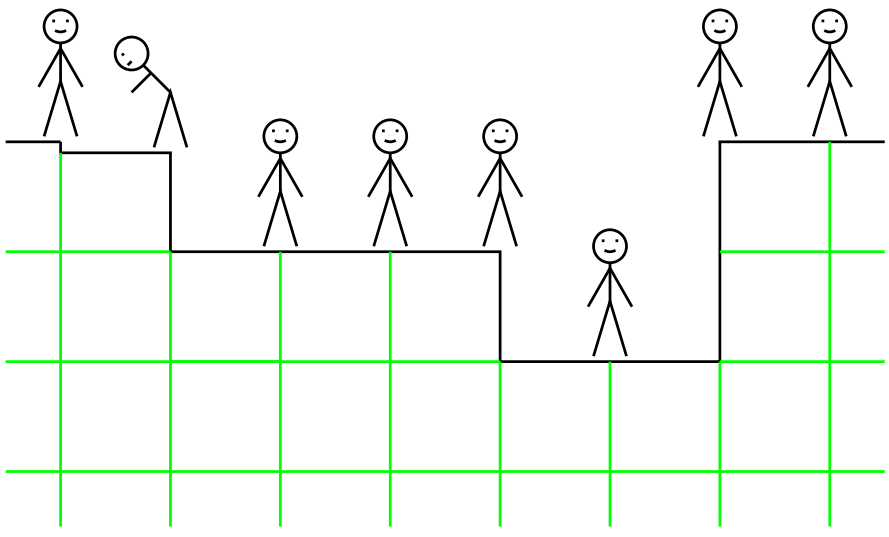
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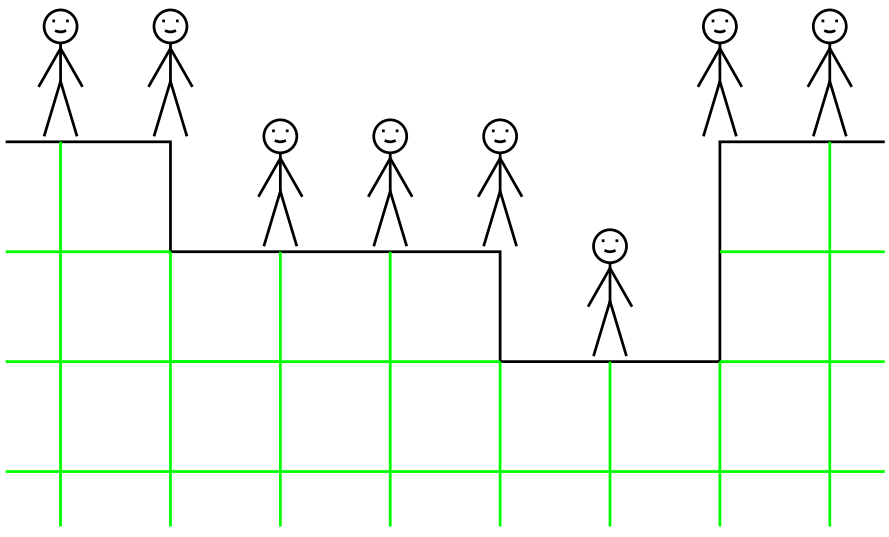
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$$\begin{pmatrix} \omega_i \\ \omega_{i+1} \end{pmatrix} \to \begin{pmatrix} \omega_i + 1 \\ \omega_{i+1} - 1 \end{pmatrix} \quad \text{with rate } q(\omega_i, \omega_{i+1}), \text{ where}$$

- p and q are such that they keep the state space (ASEP, ZRP),
- p is non-decreasing in the first, non-increasing in the second variable, and q vice-versa (attractivity),
- they satisfy some regularity conditions to make sure the dynamics exists,
- they satisfy some algebraic conditions to get a product stationary distribution for the process.

Stationarity

Under these conditions, for every *feasible* density ϱ,

$$(\omega_i)_{i\in\mathbb{Z}} \sim \text{i.i.d. } \mu^\theta =: \underline{\mu}^\theta$$

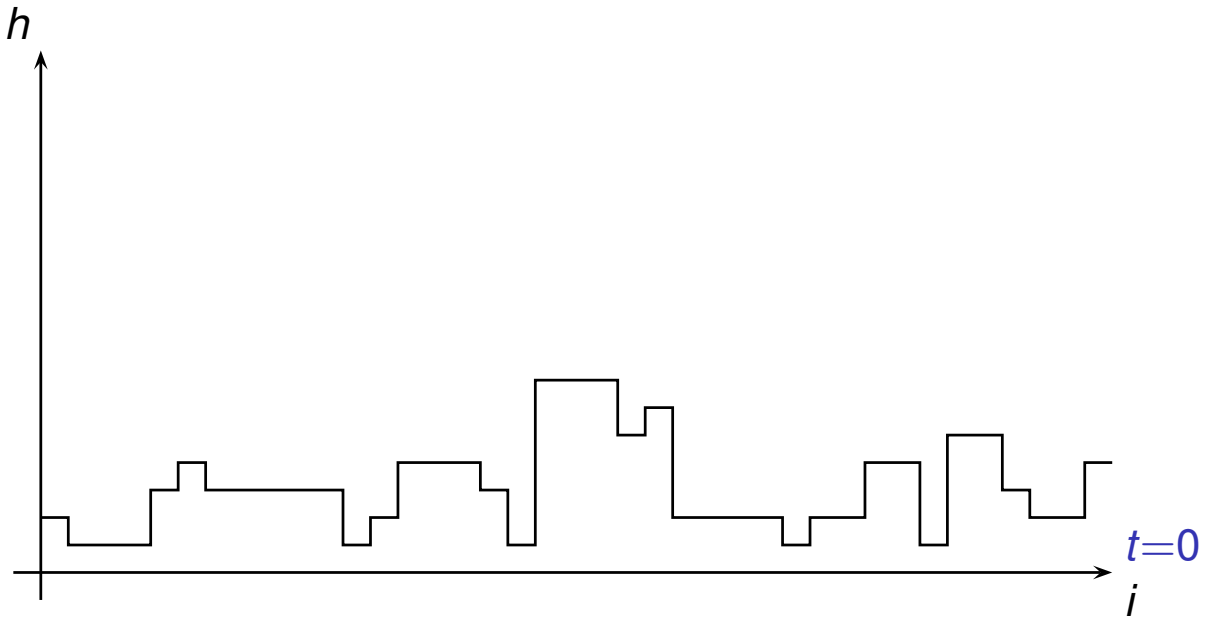
is stationary with

$$\mu^\theta(z) = \frac{\mathrm{e}^{\theta z}}{f!(z)\cdot Z(\theta)}.$$

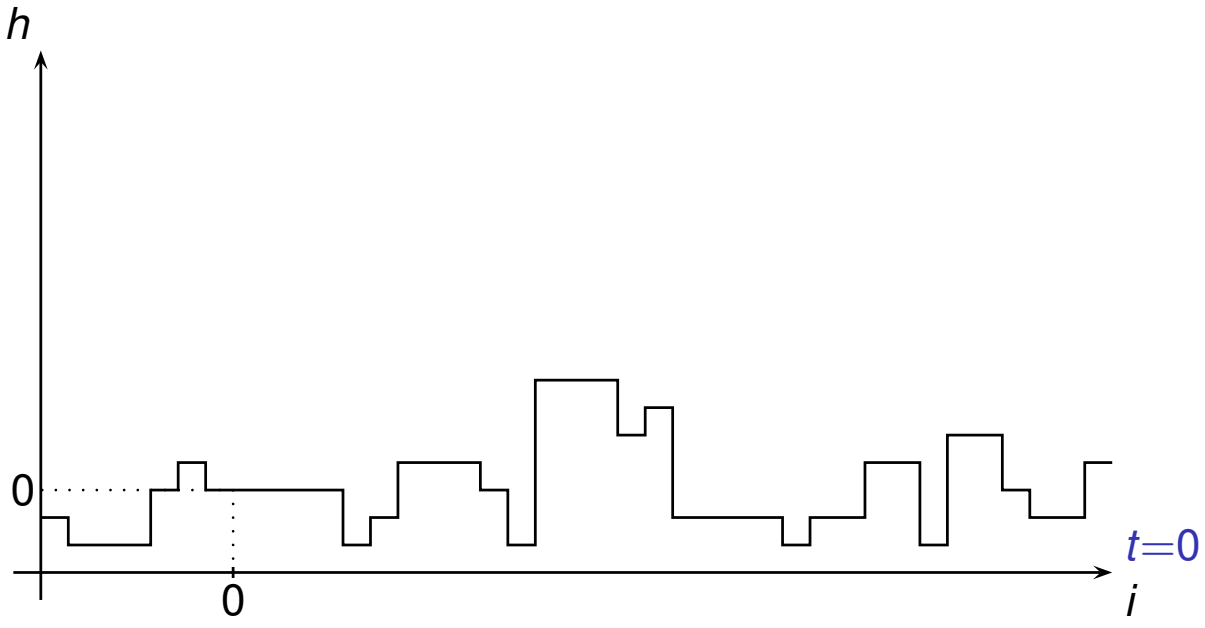
Here

$$\varrho = \varrho(\theta) = \mathbb{E}^\theta\, \omega_i = \sum_z z\mu^\theta(z).$$

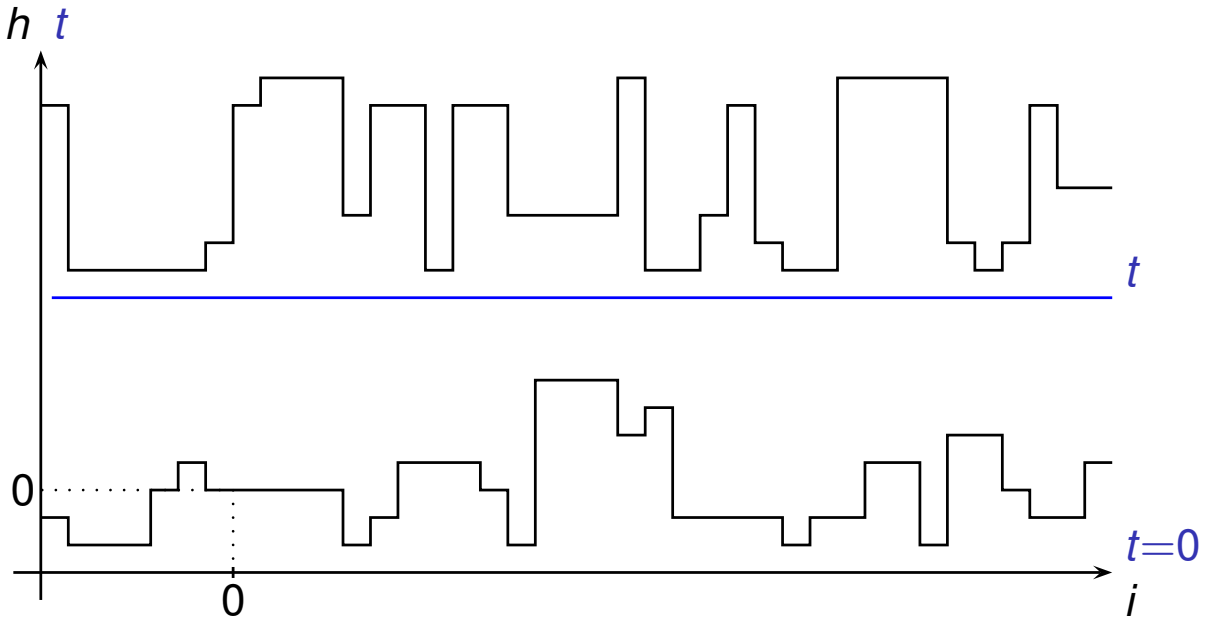
Integrated particle current



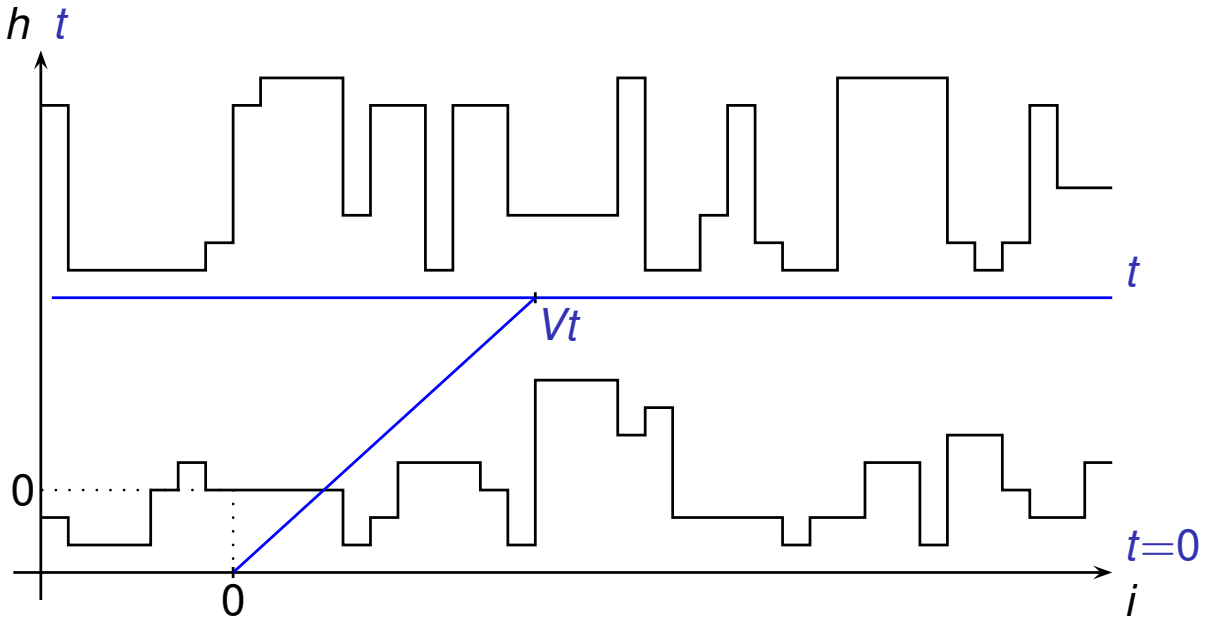
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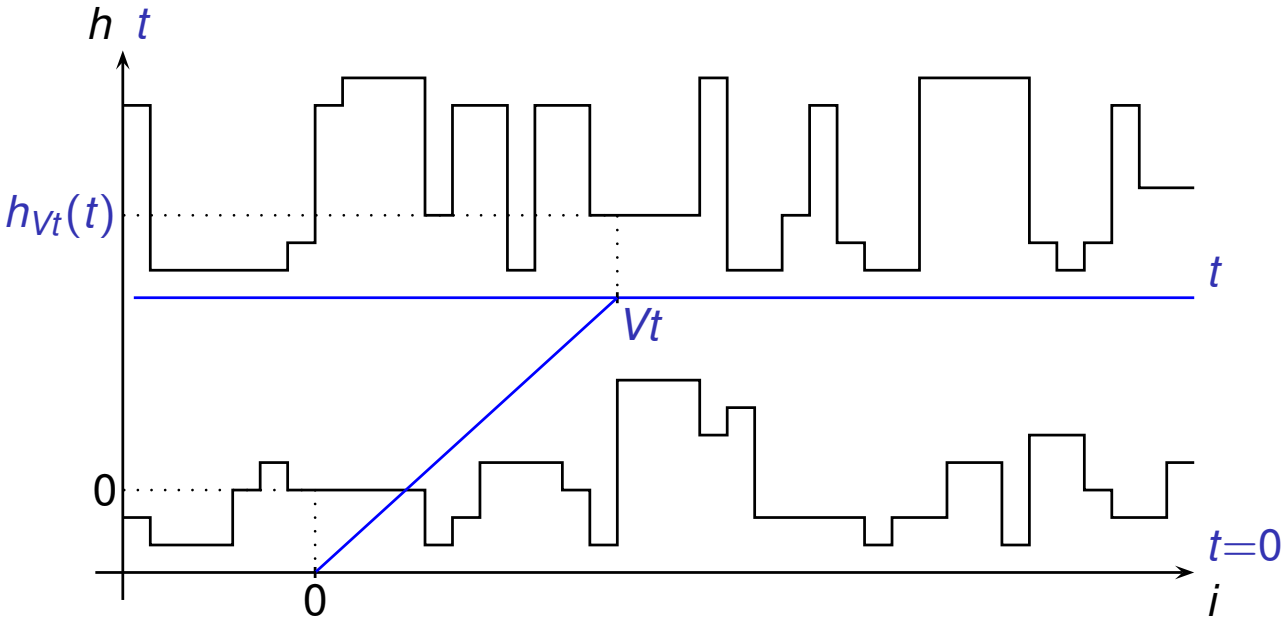
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$h_{Vt}(t)$ = height as seen by a moving observer of velocity V.

= net number of particles passing the window $s \mapsto Vs$.

(Remember: particle current=change in height.)

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Martingales (as started by Bálint some 26 years ago)

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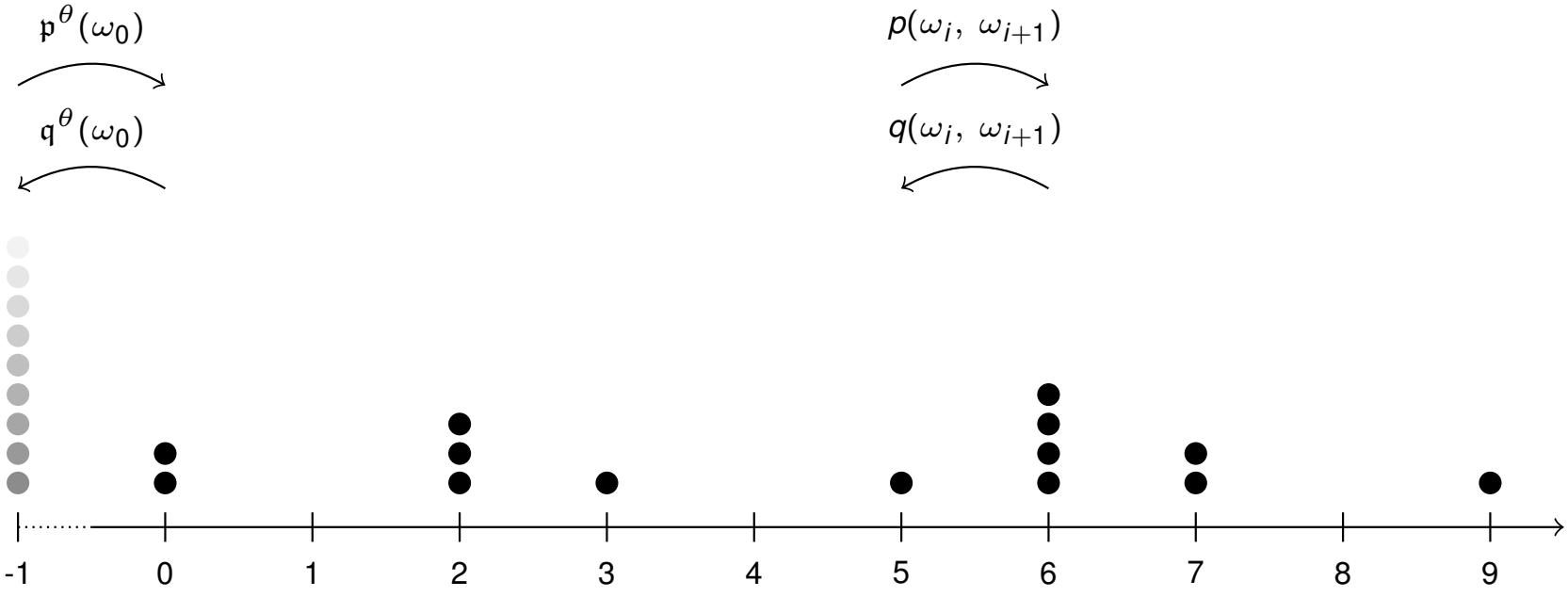
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- Basis for KPZ-scaling: $\mathbb{V}\mathrm{ar}\, h_{Ct}(t) \sim t^{2/3}$.

Boundaries



Replace ω_{-1} by a *ghost site*, and let

$$\mathfrak{p}(\omega_0) := \sum_z p(z, \omega_0) \cdot \mu^\theta(z), \qquad \mathfrak{q}(\omega_0) := \sum_z q(z, \omega_0) \cdot \mu^\theta(z).$$

$\rightsquigarrow$ With this boundary, $\underline{\mu}^\theta$ is still stationary.

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since $h_0(t) - h_0(0) \sim \text{Poi}(te^{\theta})$.

EJS for TAZRP (B., E. Emrah, J. Jay, B. Landon, E. Russell, P. Sosoe)

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But what if no explicit boundary is available?

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A few identities without boundary ($\underline{\mu}^{\gamma}$ between 1 and x):

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Thank you.